Process Characterization Report

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCR-006

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

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Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
AICc	Akaike information criterion with the small-sample correction
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CE-SDS	capillary electrophoresis with sodium dodecyl sulfate
CEX	cation exchange chromatography
CI	confidence interval
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DCS	distributed control system
DoE	design of experiments
DS	drug substance
GMP	good manufacturing practice
GPP	general process parameter
HMW	high molecular weight species
HPLC	high performance liquid chromatography
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
KPP	key process parameter
LMW	low molecular weight species
LRF	log reduction factor
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
OFAT	one factor at a time
PAR	proven acceptable range

Abbreviation	Meaning
QA	Quality Assurance
RSM	response surface methodology
SD	standard deviation
SDM	scale-down model
SEC	size exclusion chromatography
SOP	standard operating procedure
TCID50	50 % tissue culture infectious dose
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

Executive summary

Low-pH viral inactivation is Step 6 of the A-Mab drug substance process. The Protein A eluate is held at acid pH for a set time and then neutralized. The hold inactivates enveloped viruses. The drug substance requires a cumulative clearance of xenotropic murine leukemia virus (XMuLV) of at least 16.7 $\log_{10}$, which three steps claim together, and the claim of this step is 6.60 $\log_{10}$ of it. The same conditions can add aggregate, fragments and acidic variants to the pool. This report presents the Stage 1 characterization of the step under its plan PCP-006 and the master plan PCMP-001 (U.S. Food and Drug Administration 2011). The studies were run in a scale-down model of the step, qualified for aggregate and acidic variants, and in a small-scale spiking model, at the development site in Cambridge, MA.

Inactivation pH, hold time and temperature were studied in a screening design of 11 runs and a response-surface design of 18 runs. A-Mab concentration was studied one factor at a time. These characterization studies ran from 2026-01-07 to 2026-04-17. Five further studies were made from 2026-09-19 to 2026-09-29, during the Version 2.0 re-analysis and after the results of the designs were known. They are registered as discrepancies from the plan (Section 13). The findings are the following.

F1. At the normal operating ranges (NORs) against which the studies were planned, the XMuLV claim was not met. The lowest mean predicted log reduction was 6.16 $\log_{10}$, at pH 3.6, 60 min and 19 °C, against the claim of 6.60 $\log_{10}$. The pH set-point and NOR were therefore moved to pH 3.4 and pH 3.3–3.5, the hold-time NOR was narrowed to 100–120 min with its set-point at 110 min, and the temperature NOR was narrowed to 20–23 °C. Across the NOR in force the lowest mean predicted log reduction is 6.77 $\log_{10}$, so the claim is met there.

F2. Aggregate, fragments and acidic variants meet their intermediate acceptance criteria across the NOR in force. The upper limit of the 95 % prediction interval of a single batch of aggregate is at most 2.10 % HMW at manufacturing scale, against 7.02 %. Fragments and acidic variants have no adequate model and are judged on their measured results, 0.87 to 1.13 % and 20.2 to 20.6 % respectively. The 33 acidic variant results of the designs and the first concentration study show no scatter at a given hold time, so no adequacy test can be applied to that output.

F3. The selected models of aggregate and of the XMuLV log reduction meet all four model-adequacy criteria of PCMP-001 §7.2. Five runs carry a Cook’s distance above the cut-off at which PCP-006 §7.1 calls for an examination. Each was deleted in turn, the model selected again on the remaining 17 runs and the output judged over the NOR again. Under every one of those models the NOR in force lies inside the proven acceptable range (PAR) of each factor, the claim is met over the NOR, aggregate keeps its margin to its criterion, the move of the pH NOR remains necessary, and

inactivation pH stays a critical process parameter, hold time and temperature well-controlled critical process parameters and A-Mab concentration a general process parameter. With run 5 deleted the temperature term leaves the XMuLV model and the XMuLV ranges widen. Temperature raises aggregate by 0.204 to 0.226 percentage points in every model after a deletion.

F4. The PARs are pH 3.2–3.61, 84–180 min and 17.2–25 °C. One edge of each is the setting at which the mean predicted log reduction falls to the claim, and the other is the end of the range studied. The NOR in force lies 0.11 pH units inside the upper pH edge, 16 min inside the lower hold-time edge and 2.8 °C inside the lower temperature edge, and 0.10 pH units, 60 min and 2.0 °C inside the three edges that are ends of the range studied.

F5. Inactivation pH was classified as a critical process parameter (CPP). Hold time and temperature were classified as well-controlled critical process parameters (WC-CPPs), and A-Mab concentration as a general process parameter (GPP). For XMuLV clearance the concentration study bounds the effect loosely, with four runs, so the GPP class rests for that attribute on how the hold acts. The acid denatures the envelope of the virus, and the antibody around it neither shields the envelope nor takes part in that reaction.

F6. Within the NOR as characterized the step adds at most 0.56 percentage points of aggregate at the one-sided 95 % prediction bound of a single batch, against the platform value of 0.70. That bound moved the aggregate criterion of the production bioreactor output from 6.32 to 6.46 % HMW. At a hold of 180 min, outside the NOR, the step added up to 0.84 points. The platform values for fragments and acidic variants stand.

F7. At the new pH set-point aggregate and fragments showed no effect of pH. Acidic variants were 0.38 percentage points lower at pH 3.4 than at pH 3.5, a significant difference after adjustment ($p = 0.025$) that was already present before the hold began and that is 0.38 points against a criterion 22.0 points wide. Cation exchange cleared aggregate and fragments from that material as the criteria assume. At the low end of the pH PAR, with hold time and temperature at the upper ends of their NORs, product quality met every criterion of the step and cation exchange cleared aggregate as required. For fragments every run cleared more than the required clearance factor, but the lower confidence bound of the three runs fell below it, so the rule fixed before the runs was not met. That was accepted on its risk, on the grounds PD-006-03 records and Section 7.5.3 gives.

F8. The neutralized pool met its criteria after 7 days at 21 °C and 25 °C.

F9. The capability index (Cpk) of cumulative XMuLV clearance is 1.70, which is above the target of 1.33. Acidic variants have the lowest Cpk of the attributes this step changes, 1.05, which lies between the minimum of 1.00 and the target of 1.33. The lower end of the criterion limits it. The attribute is formed in the production bioreactor, the simulated drug substance mean is 22.4 % with a batch standard deviation of 1.4 points, and the lower limit of the drug substance criterion is 18.0 %. The low-pH hold raises acidic variants, so it moves them away from the limit that binds.

F10. The record of execution holds two study events. A hold-time record discrepancy of 4 min between the control system and the manual log (DEV-006-01) changes the predicted log reduction by 0.040 $\log_{10}$ and the predicted aggregate by 0.016 percentage points. Cytotoxicity of the infectivity assay at low dilution (DEV-006-02) raises the lowest titre the assay can report, which can only make a computed log reduction smaller. The runs of both events were retained in the analysis. Eight discrepancies from the plan are registered, five of them studies made during the re-analysis (Section 13).

The ranges, classes and controls of this report roll up into the Process Characterization Master Report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose drug substance is made by Novacyte Biologics. It is purified by the Novacyte platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and ultrafiltration and diafiltration (CMC Biotech Working Group 2009). The process is transferred under the Process Transfer Plan PTP-001 from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI, where it runs at the 15,000 L scale.

Low-pH viral inactivation follows Protein A capture. The Protein A eluate leaves the column at low pH. It is adjusted to the inactivation pH, held for the hold time at the set temperature and then neutralized with base. The neutralized pool is the load of cation exchange. The step is operated under its standard operating procedure, SOP-2009.

The step contributes one of the three claims that meet the cumulative requirement for clearance of the enveloped model virus XMuLV. The other two claims are those of anion exchange and virus filtration. The step can also add aggregate, fragments and acidic variants, which are formed upstream, so each of them has an intermediate acceptance criterion at the output of this step (Section 7.1). No claim is made for the parvovirus minute virus of mice (MVM), which has no lipid envelope for low pH to disrupt. No clearance of host cell protein is claimed for the step.

This report is paired with the plan PCP-006, under which the studies were run. Its results roll up into PCMR-001 with the reports of the other steps. Table 1.1 lists the related documents.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-006	Process Characterization Plan (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Document ID	Title	Relationship
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.2 Objectives

PCP-006 §2 set six objectives for the studies, and this report answers each of them.

1. To quantify the effects of inactivation pH, hold time and temperature, and of their interactions, on XMuLV clearance, aggregate, fragments and acidic variants (Section 5).
2. To fit response-surface models that predict these outputs over the range studied, and to test each model against the adequacy criteria of PCMP-001 §7.2 (Section 5.3).
3. To determine, for each attribute that has a criterion at this step, the PAR of inactivation pH, hold time and temperature, and to establish whether the NOR of each lies within it (Section 7.4).
4. To verify the platform worst cases of this step on which the criteria upstream of it rely (Section 7.7).
5. To assess the effect of A-Mab concentration on aggregate, acidic variants and XMuLV clearance one factor at a time, and to support its range against the criteria for aggregate and acidic variants (Section 7.6).
6. To provide the evidence for the classification of each parameter under SOP-4001, and for the contribution of the step to the control strategy in PCMR-001 (Section 9, Section 10).

1.3 Regulatory and scientific basis

The studies form the process design stage of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). They follow the enhanced approach of ICH Q8(R2) (International Council for Harmonisation 2009) and ICH Q11 (International Council for Harmonisation 2012), in which prior knowledge and risk assessment identify the parameters that may affect each attribute and designed experiments then measure and bound those effects. Criticality is treated as a continuum, as PCMP-001 §3 states it after ICH Q8(R2). The parameters were ranked under ICH Q9 in the risk assessment RA-001 before characterization began (International Council for Harmonisation 2023b).

Viral safety follows ICH Q5A(R2) (International Council for Harmonisation 2023a). Each claimed step is studied in its own small-scale spiking model, a laboratory model

of the step into which a known titre of the model virus is added, and the cumulative claim is built from the independent log reductions of low-pH inactivation, anion exchange and virus filtration. The process model, the attribute framework and the decision logic for parameter classes come from the A-Mab case study (CMC Biotech Working Group 2009). The designs and their analysis follow standard response-surface methodology (Montgomery 2017), with the Box-Cox transformation (Box and Cox 1964) and model selection by the small-sample information criterion AICc (Hurvich and Tsai 1989).

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 is issued under the QA investigation INV-001, which QA opened on 2026-08-03 after its review of PCMR-001 Version 1.0 for readiness of the Biologics License Application (BLA) and closed on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the PAR analysis were in no approved protocol and had been set after the data were seen (INV-001 finding 1). A second investigation, INV-002, opened on 2026-08-31 and closed on 2026-09-28 with the classification minor, applied the model-adequacy criteria of PCMP-001 §7.2 to every response-surface model with a criterion. This revision makes the following changes.

1. Each output of the step is judged against the intermediate acceptance criterion that PCMP-001 Version 2.0 derives for it, and PCP-006 §7.2 states (Section 7.1). Table 1.2 compares those criteria with the limits the Version 1.0 report applied. The Version 1.0 limits were applied at analysis and appear in no approved plan (INV-001 finding 1). The Version 2.0 aggregate criterion of 7.02 % HMW is wider than the Version 1.0 limit of 2.17 % and stricter than its own a priori value of 10.0 %, because cation exchange cleared less aggregate within its NOR than the platform worst case assumed, so the verification there tightened the criterion (Section 7.1). The XMuLV criterion of 6.60 $\log_{10}$ is the platform claim of the step and is stricter than the Version 1.0 value of 4.93 $\log_{10}$, which was the cumulative requirement minus the nominal clearance of the other steps in this campaign.
2. Fragments are now an output of the step. They are judged on results by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS), following the method decision AMD-001 of 2026-03-20. Size exclusion chromatography (SEC) reads fragments high in the Protein A eluate and in the neutralized pool, because part of the host cell protein in those matrices elutes in its low molecular weight window (INV-001 finding 6).
3. Every result judged against a criterion for aggregate or acidic variants is stated at manufacturing scale, by subtracting the offset of the scale-down model for that output (Section 3.1).
4. Every response-surface output is analysed by model selection, as corrective action CA3 of INV-002 requires (Section 3.5). PCP-006 §5.5 prescribed the full quadratic model, and the change is registered as the model-selection discrepancy PD-006-04 (Section 13). The selected models of aggregate and of XMuLV

clearance meet all four model-adequacy criteria of PCMP-001 §7.2. Acidic variants and fragments have no adequate model. Version 1.0 fitted the full quadratic model of every output and set ranges from all of them.

5. XMuLV clearance is judged on the mean predicted log reduction at the least favourable combination of the NOR, which is the worst case PCMP-001 §6.4 sets for viral clearance (Section 3.5). Version 1.0 judged it on a single batch, as it judged the quality attributes.
6. The pH set-point and NOR, the hold-time set-point and NOR, and the temperature NOR are changed by this revision (F1, Section 7.5). The studies were planned against the Version 1.0 values and run around them.
7. The execution window of every study is recorded (Table 4.3). The Version 1.0 record gave no execution dates, so the equipment tables of the package appeared to cite expired calibrations (INV-001 finding 5).
8. Five studies made during the re-analysis are reported here and registered as discrepancies from the plan. They are the hold at the new pH set-point against the former one (S-VI-OFAT), the cation exchange verification runs (S-CEX-VER), the confirmation runs at the low end of the pH PAR with hold time and temperature at the upper ends of their NORs (S-VI-WC), the hold study of the neutralized pool (S-VI-HLD) and the confirmation study of XMuLV clearance across the A-Mab concentration range (S-VI-UNI2) (Section 13).
9. Every departure from the plan is recorded as a discrepancy from the plan and every event of execution as a study event. PCP-006 calls a departure from the plan a deviation. A process characterization study is not manufacture under good manufacturing practice (GMP), so this report records no deviation, and the departures found in the re-analysis are discrepancies from the plan (Section 13).

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, low-pH viral inactivation.

Attribute	Version 1.0[a]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Acidic variants (%)	none[b]	–	18.0–40.0	Drug substance
Aggregates, HMW (%)	≤ 2.17	R1, margin 6.0	≤ 7.02	Carried back, tightened[c] from ≤ 10.0
Fragments, LMW (%)	not assessed[d]	–	≤ 2.60	Carried back
XMuLV clearance (log10)	≥ 4.93	R3	≥ 6.60	Platform claim

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[c] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[d] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.
[R3] Version 1.0 rule: requirement − nominal of other steps.

Two Version 1.0 conclusions moved. The Version 1.0 analysis found the hold-time PAR for aggregate to end at 144 min, because it judged aggregate against a limit

three times stricter than the criterion now in force. Under the Version 2.0 criterion that PAR covers the whole range studied. The Version 1.0 analysis also found XMuLV clearance acceptable across the whole NOR as characterized, against its criterion of 4.93 $\log_{10}$, with a pH PAR ending at pH 3.96. Against the platform claim and on the selected model, no XMuLV PAR exists at the NOR as characterized, because the lowest mean prediction at the former pH set-point of pH 3.5 is already below the claim (Section 7.5). The design space moved with the criteria. Version 1.0 drew it on the mean prediction of every output, and 79.9 % of the region studied met its criteria, bounded by aggregate. Version 2.0 draws it on both ends of the 95 % prediction interval of a single batch for a quality attribute and on the mean prediction for viral clearance, and 63.9 % of the region studied meets the criteria, bounded by XMuLV clearance (Section 6). One conclusion did not move. Aggregate and acidic variants met their criteria across the NOR under both revisions.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The low-pH hold belongs to the Novacyte downstream platform and was used to manufacture three earlier humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab. The process conditions of the step have remained essentially unchanged for those products and through the development of A-Mab (CMC Biotech Working Group 2009). That history supplies the platform claim for XMuLV clearance and the largest increases in aggregate, fragments and acidic variants the step has shown on the platform. The intermediate acceptance criteria of Section 7.1 are built on those values, and Section 7.7 checks them against the A-Mab data.

Low pH inactivates an enveloped virus by denaturing the glycoproteins of its envelope and by disrupting the envelope itself. The virus needs those glycoproteins to bind to a cell and to fuse with its membrane. Inactivation is commonly described by first-order kinetics, in which the log reduction grows with the hold time at a rate that rises as the pH falls and as the temperature rises. On that description a fall in pH was expected to raise the log reduction most steeply near the operating range, because the envelope glycoproteins denature much faster with each small fall in pH there. A longer hold and a warmer hold were each expected to raise the log reduction, and the effect of the hold time was expected to be larger at low pH. Section 5.4 compares that description with the surface the data give.

The acid acts on the antibody as well as on the virus. At acid pH the second constant domain of the heavy chain is partially unfolded, which exposes hydrophobic surface. Molecules in that state associate, so aggregate forms for as long as the pool is held at low pH, and faster in a warmer pool, because both the unfolding and the association speed up with temperature. Aggregate formed in the hold is carried into cation exchange, the step that removes it. Time at low pH also adds acidic variants, through direct acid-catalysed hydrolysis of side-chain amides, and a small amount of acid-catalysed cleavage in the hinge region, which adds fragments. Over a hold of the length studied, both increases were expected to be small, the acidic variants against the level formed in the production bioreactor and the fragments against the level the load already carries.

The step was not expected to change the glycan attributes. Those are formed in the production bioreactor, and no downstream step changes them (PCMP-001 §3). The criteria also assume no change in host cell protein across the step.

2.2 Quality attributes in scope

Table 2.1 lists the attributes in scope, with their drug substance criteria and their criticality from the register of critical quality attributes (CQAs) in PCMP-001 §3. Criticality was assigned with Tool #1 of the A-Mab case study, as the product of impact and uncertainty (CMC Biotech Working Group 2009). XMuLV clearance was assigned very high criticality by rule, because a viral contamination would harm the patient directly whatever its score. Attributes of high or very high criticality are called critical.

Table 2.1: Quality attributes in scope of this report, from the drug substance CQA register.

Attribute	Category	Drug substance criterion	Criticality	Tool #1 score
Viral clearance — XMuLV (enveloped)	viral safety	≥ 16.7 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60
Fragments (LMW, clipped species)	size variant	≤ 2 % (nrCE-SDS)	L	12
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Tool #1, the score of impact multiplied by uncertainty (A-Mab case study). VH, very high. H, high. L, low. VL, very low. Viral clearance is very high by rule, whatever its score. CEX/icIEF, the two methods the register names for acidic variants, cation exchange chromatography and imaged capillary isoelectric focusing. This campaign measures them by icIEF.

Aggregate and acidic variants are formed in the production bioreactor, and the register assigns fragments to harvest and clarification, where proteases released from lysed cells clip the antibody. The hold can add to each of those three, so each is measured at the output of this step and judged against an intermediate acceptance criterion (Section 7.1).

2.3 Risk-based prioritization and parameter selection

RA-001 assessed every parameter of the step before characterization began. Each parameter was ranked on its potential impact on the CQAs and on its potential to interact with other parameters, as ICH Q9 describes (International Council for Harmonisation 2023b). RA-001 gives the scores and the rationale for each assignment.

Inactivation pH, hold time and temperature were assigned to multivariate study. A lower pH, a longer hold and a warmer hold were each expected to raise the log reduction of XMuLV, and a longer or warmer hold to raise the aggregate that forms. The three were also expected to interact, because on the first-order description the log reduction is the product of a rate that depends on pH and temperature and the time over which that rate acts. A design that varies the three together was therefore needed to estimate the clearance and the aggregate at their combinations.

A-Mab concentration was assigned to univariate study. A higher antibody concentration raises the rate at which partially unfolded molecules meet, so aggregate was expected to form faster in a more concentrated pool. No effect of the antibody concentration on the inactivation of the virus was expected. The acid denatures the envelope of the virus, and the antibody neither shields the envelope nor takes part in that reaction. The concentration is also a property of the Protein A pool rather than a setting of the hold, so it was assessed one factor at a time.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model holds the Protein A eluate in a temperature-controlled vessel at laboratory scale. The eluate is adjusted to the inactivation pH, held and neutralized in that vessel as in manufacture. The parameters of the step do not depend on scale, so inactivation pH, hold time, temperature and A-Mab concentration were set to the same values as in manufacture, and the additions of acid and base were scaled with the volume of the pool. The model was operated under SOP-2009 and qualified under SOP-1001.

The qualification study S-VI-SDM ran from 2026-01-07 to 2026-01-16, before the first characterization run. It compared 3 runs of the scale-down model at set-point with 5 manufacturing-scale batches, for aggregate and acidic variants in the neutralized pool. Table 3.1 gives the offset of each output, the scale-down mean minus the manufacturing mean, with its 90 % confidence interval. Its last two columns give the qualification. The first of them shows whether the output meets its criterion at every combination of NOR low, set-point and NOR high with the offset at the low end of its interval, at its estimate and at the high end.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, low-pH viral inactivation.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Acidic variants (%)	3	5	20.69	20.41	0.27	−0.29 to 0.84	Yes / Yes / Yes	Yes
Aggregates, HMW (%)			1.94	1.92	0.02	−0.01 to 0.05	Yes / Yes / Yes	Yes

SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

Aggregate is 0.02 percentage points higher in the scale-down model and acidic variants 0.27 points higher, and neither interval excludes zero. Every result of these two outputs is translated to manufacturing scale by subtracting the offset of that output before it is compared with a criterion. The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end

of its interval (PCMP-001 §7.1). For aggregate that conclusion is the upper end of the single-run prediction interval of the selected model over the NOR against 7.02 % HMW. For acidic variants, which have no adequate model, it is every measured result of the designs against 18.0 to 40.0 %. Both conclusions hold at all three values of the offset, so the model is qualified for aggregate and for acidic variants.

The qualification compared neither fragments nor XMuLV clearance at both scales, so the offset of each is unknown rather than zero. These two outputs are judged as the laboratory model gives them, with no translation to manufacturing scale. For fragments the representativeness of the model rests on the mechanism of the step, as PCP-006 §5.1 states it. Cleavage in the hinge region at low pH depends on pH, time and temperature, which the scale-down model sets to the manufacturing values, and not on the size of the vessel. The fragment results are far from their criterion, which bounds what the assumption can cost. The highest single result of the designs is 1.13 %, which leaves 1.47 points to the criterion of 2.60 %. The assumption is stated as a limitation in Section 11.

Virus cannot be introduced into the manufacturing process, so XMuLV clearance was measured in the spiking model of the step (International Council for Harmonisation 2023a). PCP-006 §5.1 sets out the spiking model. It is the scale-down model, run on the same Protein A eluate and at the same inactivation pH, hold time and temperature. The XMuLV spike is added after the pool has been acidified and before the hold begins, at the volume ratio fixed in SOP-2009, which is small enough that the spike changes neither the pH nor the antibody concentration of the pool. The titre of the spiked pool is measured at the start of the hold and at the end. The virus load of a sample is its titre multiplied by the volume of the pool, and the log reduction factor (LRF) is the difference between the $\log_{10}$ load at the start and the $\log_{10}$ load at the end. Each run of a design is made twice, once spiked for the titres and once unspiked for the quality attributes. The inactivation rate depends on the pH, the temperature and the matrix of the pool, and not on its volume. The spiking runs use Protein A eluate produced at the set-points of the Protein A step, so the matrix is that of manufacture, and the runs of the designs held the A-Mab concentration at its set-point. The small-scale result therefore represents the manufacturing step at that concentration, provided the whole pool reaches the inactivation pH.

The centre points of each design are replicate runs at the design centre. They give the pure error against which lack of fit is tested (Section 5.1).

3.2 Operation

The runs followed the operation of PCP-006 §5.2. Each run starts from Protein A eluate produced at the set-points of the Protein A step, which is characterized under PCR-005. The eluate is adjusted to the inactivation pH of the run and brought to its temperature, and the hold time is counted from the moment the inactivation pH is reached. At the end of the hold the pool is neutralized with base and filtered. Neutralization stops the inactivation and slows the reactions that the acid drives on the antibody. In both designs the A-Mab concentration was set to its set-point of 20 g/L.

The input pool was not tested, so that is the value set and not a measured result (PD-006-08). In the two concentration studies the concentration is set to each planned value before the pH is adjusted.

The acid and base solutions and the other inputs of the step are prepared to their platform specifications. They are controlled for identity and quality and were not characterized.

3.3 Analytical methods

Each output was measured by a validated method under its own procedure. Table 3.2 lists the procedures and method validations this report cites. XMuLV titre was measured as the 50 % tissue culture infectious dose (TCID50) under AMV-3017, aggregate by SEC under AMV-3011, acidic variants by icIEF under AMV-3013 and fragments by nrCE-SDS under AMV-3020. Appendix D summarises the methods and their reportable units.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2009	Operation of the Low-pH Viral Inactivation Step
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3013	Charge Variants (icIEF)
	AMV-3020	Purity (non-reduced CE-SDS)

From 2026-03-20, nrCE-SDS is the method for fragments wherever fragments are measured (PCMP-001 §7.8). Before that date fragments were measured by SEC. SEC remains the method for aggregate, which is read in a window the host cell protein of these matrices does not reach. Table 3.3 shows which method measured which attribute in which design.

Table 3.3: Methods used for size variants in the designed studies, low-pH viral inactivation.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	
Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]

[a)] In the Protein A and low-pH viral inactivation pools part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained samples of the response-surface design, tested 2026-04-06 to 2026-04-17.

Both designs ran before the method decision and measured fragments by SEC. The screening samples were not retained, so the screening design has SEC fragment results only. Those results are kept in the record and are used for no decision in this report. The retained samples of the response-surface design were tested by nrCE-SDS in the re-test study S-VI-RET, from 2026-04-06 to 2026-04-17, and the fragment results of the designs are those 18 results. Of the studies made during the re-analysis, S-VI-HLD, S-VI-OFAT and S-VI-WC measured fragments by nrCE-SDS, and S-VI-UNI2 measured none. The qualification study and the two concentration studies measured no fragments, and no fragment result is claimed for any of them. The effects on fragments therefore rest on the response-surface design (Section 5.3), and the range of the A-Mab concentration is supported for aggregate and acidic variants but not for fragments (Section 7.6).

3.4 Sampling plan

PCP-006 §5.4 sets out three samples from the spiked run and two from the unspiked run of each design run. The spiked run gives the input pool, the spiked load at the start of the hold and the spiked pool at the end, so the two titres of one LRF come from the same run. The unspiked run gives the input pool and the neutralized pool, sampled after neutralization and filtration, which is the level the step delivers to cation exchange. The record of both designs holds an LRF and a result for aggregate, acidic variants and fragments in the neutralized pool for every run, 11 screening runs and 18 response-surface runs.

The input pool, to be sampled before acidification, was to give the level of aggregate, acidic variants and A-Mab concentration that the step starts from. It was not tested in either design, which is registered as the input-pool discrepancy PD-006-08 (Section 13). The increase across the step that the verification of the platform worst cases needs is therefore taken against the aggregate level of the load in the process model at set-point, 1.55 % HMW (Section 7.7).

3.5 Statistical methods

Each factor is coded −1 at the low end of its range studied, +1 at the high end and 0 at the midpoint (Montgomery 2017). An effect is the change in the response over the coded range of the factor, which is twice its coefficient in coded units. A model term is significant when its p-value is below $\alpha = 0.05$, the level PCMP-001 §6.2 sets. About one term in twenty with no real effect is called significant at that level. A count of terms in this report counts the intercept as one of them.

The screening data were fitted by least squares with a model of main effects and two-factor interactions, which has six terms for three factors. The screening model identifies effects. Its centre points can detect curvature but cannot attribute it to a factor, so no prediction in this report comes from it. The response-surface model is the predictive model, and every prediction, every PAR, the design space and the verification of the platform worst cases come from it.

PCP-006 §5.5 prescribed the full quadratic model of each output, which has nine terms. This report selects the model of each output instead, as corrective action CA3 of INV-002, adopted on 2026-09-15, requires for every report of the campaign. The change is registered as PD-006-04 (Section 13). The selection has three steps.

1. *The scale.* The Box-Cox profile of the full quadratic estimates the exponent λ that would make the residuals of the model best behaved, with $\lambda = 1$ the raw scale and $\lambda = 0$ the log scale (Box and Cox 1964). The log scale is used when the 95 % interval of λ contains 0 and excludes 1, which means the data support the log scale and reject the raw one.
2. *The model.* Among all 95 submodels of the full quadratic that keep strong hierarchy, an interaction only with both its main effects and a square only with its main effect, the one with the lowest AICc is taken. AICc estimates how well a model would predict new runs and penalises every term it estimates, and its small-sample correction raises that penalty when the runs are few for the number of terms, as they are in a design of 18 runs (Hurvich and Tsai 1989). The residual variance counts as one of the estimated parameters.
3. *The check.* The selected model is judged against the four adequacy criteria of PCMP-001 §7.2. The first is a significant model F-test at $\alpha = 0.05$. The second is no significant lack of fit against pure error at $\alpha = 0.05$. The third is a limit on overfitting, under which R^2 may exceed predicted R^2 by no more than 0.20. The fourth, called the criterion on residuals and influential runs in this report, asks for residuals with no pattern against the predicted values or the run order and for no run dominating the fit. PCP-006 §7.1 tests it by two regressions, one of the absolute residuals on the predicted values and one of the residuals on the order of execution, each at $\alpha = 0.05$, and by the Cook's distance of each run. A run whose Cook's distance exceeds $4/n$, which is 0.22 for 18 runs, is examined by deleting it, selecting the model again on the remaining runs and judging the output over the NOR again. The model fails the criterion where a run exceeds the 50 % quantile of the F distribution with the degrees of freedom of the model, which is a much larger distance. Predicted R^2 comes from the prediction error sum of squares, in which each run is predicted by the model fitted without it

(Montgomery 2017). An output with no adequate model sets no range and is judged on its measured results.

A model that reproduces its results exactly is never adequate. With no residual there is no pure error for the lack-of-fit test and no residual mean square for the model F-test, so the criteria of PCMP-001 §7.2 cannot be applied, and the prediction interval of such a model has zero width. The acidic variants of this step are such an output (Section 5.3).

Aggregate, fragments and acidic variants must meet their criteria in every batch, and a response-surface model predicts the mean response. Each of them is therefore judged through the 95 % prediction interval of a single run, which stands for a single batch, at every combination of NOR low, set-point and NOR high of the three factors, with the offset of the scale-down model applied. That interval is two-sided, so its upper end is a one-sided 97.5 % limit. XMuLV clearance is judged on the mean predicted LRF at the least favourable of those combinations. Viral clearance is validated at worst-case conditions, so the claim has to hold there, and this report gives that reason nowhere else. The prediction interval of a single run of the log reduction is reported beside it for information.

PCMP-001 §6.4 defines two PAR analyses per attribute and factor. The second, which PCMP-001 calls the NOR-propagated PAR, is the range of the factor over which the attribute meets its criterion while the other two factors vary within their NORs. For a quality attribute it draws 2,000 operating points of the other two factors at each of 81 grid points, each factor from a normal distribution centred on its set-point whose standard deviation is one sixth of the width of its NOR, clipped to the range studied, and compares the central 95 % of the simulated batches with the criterion. For XMuLV clearance it draws nothing. At each grid point the other two factors take every combination of their NOR low, set-point and NOR high, and the lowest mean prediction is compared with the claim. The verification of the platform worst cases uses the one-sided 95 % prediction bound of a single batch (Section 7.7). Commercial-scale capability comes from the integrated model of the train, which runs 2,000 simulated batches through every step (Section 8).

One analysis of this report is in no plan. PCP-006 sets no rule for models that fit almost as well as the selected one. Where another model lies within 2 AICc units of the selected one, that model was fitted and its conclusions compared with those of the selected model, which decides. The comparison is registered within PD-006-04 (Section 5.3.1).

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the parameters of the step with their set-points, their NORs and the ranges studied. The set-points and NORs shown are those in force, which this revision changed for three of the four parameters (Section 7.5). The values the studies were planned against are called the NOR and set-points as characterized throughout this report. The range studied is the characterized range of PCMP-001 and bounds the knowledge space within which a PAR can be claimed.

Table 4.1: Parameters of the low-pH viral inactivation step, with the set-points and NORs in force, the ranges studied and the classification of this report.

Parameter	Unit	Set-point	NOR	Range studied	Class	Study
Inactivation pH	pH	3.4	3.3–3.5	3.2–4	CPP	multivariate
Hold time	min	110	100–120	60–180	WC-CPP	multivariate
Temperature	°C	21	20–23	15–25	WC-CPP	multivariate
A-Mab concentration	g/L	20	5–35	5–35	GPP	univariate

The NOR of A-Mab concentration equals the range studied. The set-points and NORs the studies were planned against are in the table of set-points and ranges as characterized and in force, in the section on the ranges in force.

Table 4.2: Code letters of the factors in the design matrices of Appendices A and B.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

The ranges studied were set in RA-001 and PCP-006 §6.1. Inactivation pH was studied from 3.2 to 4, wider than the NOR on both sides, which shows how the step behaves near and beyond the edges of routine operation. Hold time was studied from 60 to 180 min. Its lower end was the shortest hold the NOR of PCP-006 allowed, and it is the least favourable condition for XMuLV clearance. Its upper end is 1.5 times the longest hold of that NOR and shows how aggregate and fragments grow in a long hold. Temperature was studied from 15 to 25 °C, the range a manufacturing suite delivers. The A-Mab concentration of the Protein A pool varies from 5 to 35 g/L, and its NOR equals that range.

The centre of each design is the midpoint of the ranges studied, at pH 3.6, 120 min and 20 °C. It is the set-point of none of the three factors, because the ranges are not symmetrical about the set-points. The centre points therefore estimate pure error at the design centre, and each model predicts the outputs at the set-point from the fitted surface rather than measuring them there.

4.2 Screening design

The screening design S-VI-SCR was a two-level full factorial design in inactivation pH, hold time and temperature with 3 centre points, 11 runs in all, executed in random order from 2026-01-19 to 2026-01-30. Every combination of the low and high ends of the three ranges was run, so the design estimates all three main effects and all three two-factor interactions without aliasing. Its purpose was to identify which factors and interactions act on each output. The design matrix and the results are in Appendix A.

4.3 Response-surface design

The response-surface design S-VI-RSM was a face-centred central composite design in the same three factors, with 8 factorial runs, 6 axial runs and 4 centre points, 18 runs in all, executed in random order from 2026-02-02 to 2026-02-13 (Montgomery 2017). The axial runs lie at the centres of the faces of the design cube, at the ends of each range studied, so every run stayed inside the range studied. Together with the centre points they allow a squared term to be estimated for each factor. PCP-006 §6.2 carried every factor with a detected effect into this design, and the screening had detected an effect of each of the three on XMuLV clearance (Section 5.2). Every output of the neutralized pool was measured on every run. The input pool was not tested (PD-006-08). The design matrix and the results are in Appendix B.

4.4 Univariate assessment

A-Mab concentration was assessed one factor at a time in study S-VI-UNI, from 2026-02-02 to 2026-02-06. The low-pH hold was run at 5 and 35 g/L in duplicate, with the other three parameters at their set-points as characterized, and the set-point runs of the qualification study served as the reference at set-point. PCP-006 §6.4 required aggregate, acidic variants and the log reduction of XMuLV on every run. The study measured aggregate and acidic variants only, which is registered as PD-006-07, the clearance the concentration study did not measure (Section 13). The clearance across the concentration range was measured in the confirmation study S-VI-UNI2, from 2026-09-19 to 2026-09-29, registered as PD-006-06, the clearance at the two ends of the concentration range. A study of one factor at a time cannot show whether the effect of the concentration depends on the other parameters, and the designs,

which held the concentration at its set-point, carry no information on a concentration effect.

Five studies beyond the plan were made during the Version 2.0 re-analysis and are reported here. Table 4.3 gives the execution window of every study of the step and of the cation exchange verification runs, in the order in which they started.

Table 4.3: Execution window of every study reported here.

Phase	Study	Study type	Start	End
Characterization	S-VI-SDM	Scale-down model qualification	2026-01-07	2026-01-16
	S-VI-SCR	Screening design	2026-01-19	2026-01-30
	S-VI-UNI	A-Mab concentration, aggregate and acidic variants	2026-02-02	2026-02-06
	S-VI-RSM	Response-surface design	2026-02-02	2026-02-13
	S-VI-RET	Re-test of retained samples by nrCE-SDS	2026-04-06	2026-04-17
Version 2.0 re-analysis	S-VI-UNI2	XMuLV clearance at both ends of the A-Mab concentration range	2026-09-19	2026-09-29
	S-VI-HLD	Hold of the neutralized pool	2026-09-20	2026-09-29
	S-VI-OFAT	Quality attributes at the new pH set-point	2026-09-21	2026-09-23
	S-CEX-VER	Cation exchange runs on material at the new pH set-point	2026-09-24	2026-09-25
	S-VI-WC	Confirmation runs at the low end of the pH PAR, hold time and temperature at their NOR highs	2026-09-26	2026-09-29

Start, the date of the first run or test of a study. End, the date of the last. The cation exchange verification runs are a study of the cation exchange scale-down model on material from this step.

Every characterization study closed before Version 1.0 of this report was approved on 2026-07-24, and the first of them started after PCP-006 Version 1.0 was approved on 2026-01-05. The five studies of the re-analysis ran from 2026-09-19 to 2026-09-29, after INV-001 closed on 2026-09-18. The set-points and NORs of this revision were decided on the results of the designs. The hold at the new pH set-point, S-VI-OFAT, and the cation exchange verification runs tested that set-point after it had been decided, and the confirmation runs were made after the hold-time NOR had been revised (Section 7.5).

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean, the standard deviation and the coefficient of variation of each response over the centre points of both designs. These replicates are the pure-error basis of the lack-of-fit tests in Section 5.3.

Table 5.1: Centre-point reproducibility of each response in both designs.

Design	Response	n	Mean	SD	%CV
Screening	Aggregates (HMW, %)	3	2.03	0.0324	1.60
	XMuLV LRF ($\log_{10}$)	3	7.21	0.253	3.51
	Acidic variants	3	20.7	0	0.00
Response surface	Aggregates (HMW, %)	4	2.01	0.0169	0.84
	XMuLV LRF ($\log_{10}$)	4	6.71	0.381	5.67
	Acidic variants	4	20.7	0	0.00

n, the number of centre-point runs. SD, standard deviation of the replicates. %CV, the standard deviation as a percentage of the mean. Aggregate and acidic variants in percent, XMuLV log reduction in log10.

The XMuLV log reduction is the least precise response. Its coefficient of variation is 5.7 % over the 4 centre points of the response-surface design, which is a standard deviation of 0.38 $\log_{10}$. An infectivity titre carries the variability of a cell-based assay (PCP-006 §12), and an LRF is the difference of two $\log_{10}$ virus loads, so it carries that variability twice. Aggregate reproduces to 0.8 %, a standard deviation of 0.017 percentage points. Fragments, measured by nrCE-SDS on the retained samples of the response-surface design, reproduce to a standard deviation of 0.040 points about a mean of 1.02 % over those 4 runs.

The acidic variants gave the same result at every centre point of both designs, so their coefficient of variation is zero. Across both designs and the first concentration study, 33 results in all, the attribute took one value per hold time and nothing else, 20.5 % at 60 min, 20.6 % at 90 min in the concentration study, 20.7 % at 120 min and 20.9 % at 180 min. There is no scatter about those values and no dependence on pH or temperature. The same method gave a repeatability of 0.41 percentage points on the same attribute in S-VI-OFAT (Section 7.5.1), and the four runs of the second concentration study gave four different values, so the record holds scatter on this attribute elsewhere and none in the designs. The record gives no reason for that. The consequence for the analysis is that no adequacy test can be applied to the model of this output (Section 5.3).

The centre-point mean of the XMuLV log reduction was 0.50 $\log_{10}$ higher in the screening design than in the response-surface design, which is 1.3 times the pure-error standard deviation of 0.38 $\log_{10}$ at the design centre. Section 5.2 compares the two designs at the settings they share.

5.2 Screening: factor effects

Table 5.2 gives the fit of the screening model of each response, and Table 5.3 the effect estimates. An effect is the change in the response from the low to the high end of the range studied.

Table 5.2: Screening models of the low-pH viral inactivation step.

Response	Runs	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	11	0.992	0.980	0.828	82.0	< 0.001	0.0333
XMuLV LRF ($\log_{10}$)	11	0.982	0.956	0.819	37.2	0.002	0.265
Acidic variants	11	1.000	1.000	1.000	n.a.	n.a.	n.a.

Main effects and two-factor interactions in coded units. RMSE, residual standard deviation, in the unit of the response. n.a., not applicable: the model reproduces the acidic variant results exactly, so it has no residual and no test.

Table 5.3: Screening effects on aggregate and on XMuLV log reduction, largest first.

Response	Term	Effect	Coef.	Std. err.	p-value	Sig.
Aggregates (HMW, %)	B	0.471	0.236	0.0118	< 0.001	***
	C	0.221	0.110	0.0118	< 0.001	***
	AC	0.035	0.018	0.0118	0.212	
	AB	0.028	0.014	0.0118	0.294	
	A	0.022	0.011	0.0118	0.402	
	BC	0.011	0.005	0.0118	0.674	
XMuLV LRF ($\log_{10}$)	A	−2.077	−1.038	0.0936	< 0.001	***
	B	1.723	0.862	0.0936	< 0.001	***
	C	0.636	0.318	0.0936	0.027	*
	BC	0.238	0.119	0.0936	0.272	
	AB	0.203	0.101	0.0936	0.340	
	AC	−0.194	−0.097	0.0936	0.359	

Effect, the change in the response over the coded range of the factor, which is twice the coefficient. Terms: A, inactivation pH. B, hold time. C, temperature. AB, AC and BC are the two-factor interactions. Sig., significance of the term. *, $p < 0.05$. **, $p < 0.01$. ***, $p < 0.001$.

Each of the three factors changed the XMuLV log reduction. A fall in pH across the range studied raised it by 2.08 $\log_{10}$ ($p < 0.001$), a longer hold by 1.72 $\log_{10}$ ($p <$

0.001) and a warmer hold by 0.64 $\log_{10}$ ($p = 0.027$). No interaction reached significance. All three factors were therefore carried into the response-surface design, as PCP-006 §6.2 planned for a factor with a detected effect.

Aggregate responded to hold time and temperature. A longer hold raised it by 0.471 percentage points ($p < 0.001$) and a warmer hold by 0.221 points ($p < 0.001$). The effect of pH was 0.022 points and did not reach significance ($p = 0.402$), and no interaction did either. The acidic variants gave an effect of hold time alone, of 0.4 percentage points over the range studied, and no residual scatter, so the fit statistics of that model carry no information (Table 5.2).

Fragments were measured by SEC in the screening design, and those results are biased in this matrix (Section 3). They are reported in Appendix A for completeness and are used for no decision, so the effects on fragments in this report rest on the 18 nrCE-SDS results of the response-surface design alone (Section 5.3).

The screening and the response-surface design shared their eight corner settings, and the XMuLV log reduction differed between them. At the four corners at pH 3.2 the screening runs read 0.66 $\log_{10}$ higher on average, and at the four corners at pH 4 the two designs agreed to 0.11 $\log_{10}$, which is less than the pure-error standard deviation. The pH effect was therefore larger in the screening design than in the response-surface design, 2.08 against 1.22 $\log_{10}$, and the difference is significant when the two estimates are compared against their standard errors ($p = 0.006$). For hold time and temperature no difference between the designs was detected ($p = 0.097$ and $p = 0.834$). The record holds the log reduction of each run and no titre, so whether an output titre at pH 3.2, where the clearance is largest, reached the lowest titre the assay can report cannot be checked from it. Cytotoxicity of the assay at low dilution raises that lowest titre, and a higher output titre makes a computed log reduction smaller, so DEV-006-02 is one possible contributor to the difference. The record names no cause. The response-surface model is the predictive model of this report, and it gives the smaller pH effect, so the design space and the ranges rest on the smaller estimate.

5.3 Response-surface models

Table 5.4 gives, for each output with a criterion, the fit of the full quadratic model that PCP-006 §5.5 planned, on the 18 planned runs, and in its last column the terms of the model the output ends up with under the selection of Section 3.5. The design has one block, so no selected model carries a block term.

Table 5.4: Re-evaluation of the response-surface models.

Attribute	Planned runs, full quadratic				Model used (terms)
	Runs	R^2	Pred. R^2	Gap	
Aggregates, HMW (%)	18	1.00	0.96	0.04	Hold time, temperature
XMuLV clearance (log10)	18	0.90	0.58	0.32	Inactivation pH, hold time, temperature
Acidic variants (%)	18	1.00	1.00	0.00	None[a)]
Fragments, LMW (%)	18	0.85	0.12	0.73	None[a)]

[a)] No adequate model. The output is judged on its measured results. Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, on one block of runs (INV-002).

The Box-Cox profile chose the raw scale for every output that has one. The estimates of the exponent λ are -0.98 for aggregate, 1.66 for the XMuLV log reduction and -1.80 for fragments, and each 95 % interval contains the raw scale as well as the log scale. The data therefore do not reject the raw scale, and every coefficient and prediction below is in the unit of its attribute. The acidic variants have no Box-Cox interval, because every exponent fits their results exactly.

The selected model of aggregate carries hold time and temperature. Table 5.5 gives its coefficients and Table 5.6 the partition of its sums of squares.

Table 5.5: Coefficients of the selected model of aggregate (% HMW).

Term	Coefficient	Std. err.	t	p-value
Intercept	2.02	0.00428	471	< 0.001
B	0.236	0.00574	41.2	< 0.001
C	0.107	0.00574	18.7	< 0.001

Coded units, so a coefficient is half the change over the range studied. Terms: A, inactivation pH. B, hold time. C, temperature.

Table 5.6: Model, lack-of-fit and pure-error partition of the selected model of aggregate.

Source	Sum of squares	df	Mean square	F	p-value
Model	0.673	2	0.337	1022.34	< 0.001
Residual	0.00494	15	0.000329		
Lack of fit	0.00408	12	0.00034	1.18	0.504
Pure error	0.000861	3	0.000287		

Lack of fit and pure error partition the residual. Pure error comes from the centre points.

A hold of the full range studied raises aggregate by 0.472 percentage points ($p < 0.001$) and the full temperature range by 0.215 points ($p < 0.001$). The model explains $R^2 = 0.993$ of the variation, with an adjusted R^2 of 0.992, and predicts left-out runs at predicted $R^2 = 0.988$, a difference of 0.004 against the limit of 0.20. The model F-test is significant ($p < 0.001$) and lack of fit against the pure error of the centre points is not ($p = 0.504$). The selection dropped the pH term, so every prediction of aggregate in this report comes from a model in which pH does not appear. On the full quadratic model of the planned runs the pH effect lies between −0.030 and 0.020 percentage points over the whole range studied, so these data bound a linear effect of pH on aggregate within about 0.03 points. That bound is the size of the effect the dropped term could carry.

The selected model of the XMuLV log reduction carries inactivation pH, hold time and temperature. Table 5.7 gives its coefficients and Table 5.8 its partition.

Table 5.7: Coefficients of the selected model of the XMuLV log reduction (log10).

Term	Coefficient	Std. err.	t	p-value
Intercept	6.82	0.0919	74.2	< 0.001
A	−0.612	0.123	−4.96	< 0.001
B	0.605	0.123	4.9	< 0.001
C	0.286	0.123	2.32	0.036

Coded units, so a coefficient is half the change over the range studied. Terms: A, inactivation pH. B, hold time. C, temperature.

Table 5.8: Model, lack-of-fit and pure-error partition of the selected model of the XMuLV log reduction.

Source	Sum of squares	df	Mean square	F	p-value
Model	8.21	3	2.74	18.01	< 0.001
Residual	2.13	14	0.152		
Lack of fit	1.69	11	0.154	1.06	0.546
Pure error	0.435	3	0.145		

Lack of fit and pure error partition the residual. Pure error comes from the centre points.

A rise in pH across the range studied lowers the log reduction by 1.22 $\log_{10}$ ($p < 0.001$), a hold of the full range studied raises it by 1.21 $\log_{10}$ ($p < 0.001$) and the full temperature range by 0.57 $\log_{10}$ ($p = 0.036$). The model explains $R^2 = 0.794$, with an adjusted R^2 of 0.750 and a predicted R^2 of 0.675, a difference of 0.120 between R^2 and predicted R^2, again within the limit. The model F-test is significant ($p < 0.001$) and lack of fit is not ($p = 0.546$). Its residual standard deviation is 0.39 $\log_{10}$, close to the pure-error standard deviation of the centre points, 0.38 $\log_{10}$, so the model leaves no more scatter than replicate runs at the same settings show. No interaction entered the model, and Section 5.4 bounds the interaction of pH with hold time on the full quadratic model the plan prescribed.

Acidic variants and fragments have no adequate model. For the acidic variants the full quadratic reproduces every result exactly, as Section 5.1 shows, and the criteria of PCMP-001 §7.2 cannot be applied to such a fit (Section 3.5). The model that the selection returns carries hold time and is not adequate, so it sets no range and the attribute is judged on its measured results. Those results depend on hold time alone (Section 5.4).

For fragments the selected model carries inactivation pH, hold time, inactivation pH² and hold time². It is significant ($p = 0.003$) and shows no lack of fit ($p = 0.403$), but $R^2 = 0.678$ (adjusted R^2 0.579) falls to predicted $R^2 = 0.350$, a difference of 0.328 against the limit of 0.20. The model therefore has more terms than these data support, and it sets no range. It does meet the criterion on residuals and influential runs. Neither residual regression is significant ($p = 0.162$ and $p = 0.174$), and no run reaches the F limit, although runs 4 and 12 carry a Cook's distance above the cut-off for examination. Fragments are judged on the 18 nrCE-SDS results of the response-surface design, which run from 0.87 to 1.13 %. At the rate S-VI-OFAT measured (Section 7.5.1) the hold adds about 0.10 points over the whole hold-time range studied, which is 1.6 times the repeatability of nrCE-SDS, 0.06 points, and leaves 1.47 points to the criterion.

Figure 5.1 and Figure 5.2 show the residuals of the two selected models against their predicted values and on a normal probability plot, and the predicted against the measured values.

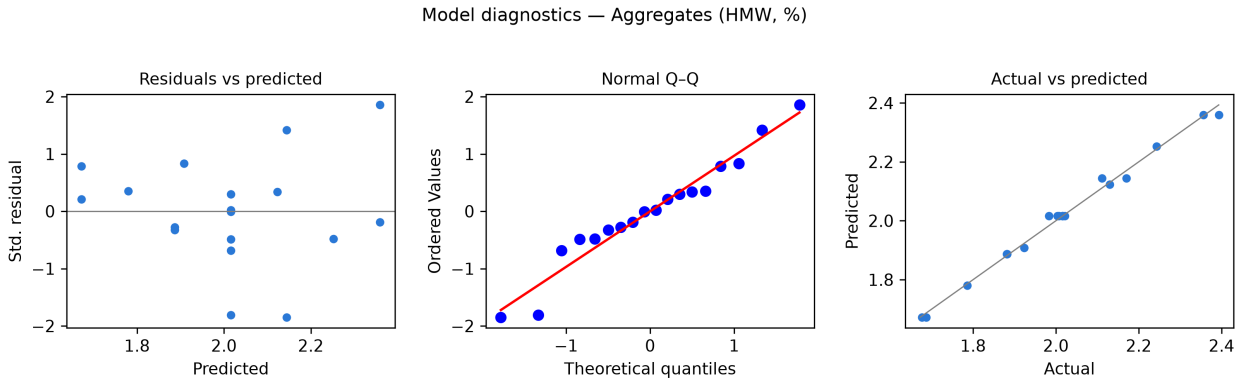


Figure 5.1: Diagnostics of the selected model of aggregate. Left, standardized residuals against predicted values. Middle, normal probability plot of the residuals. Right, predicted against measured values.

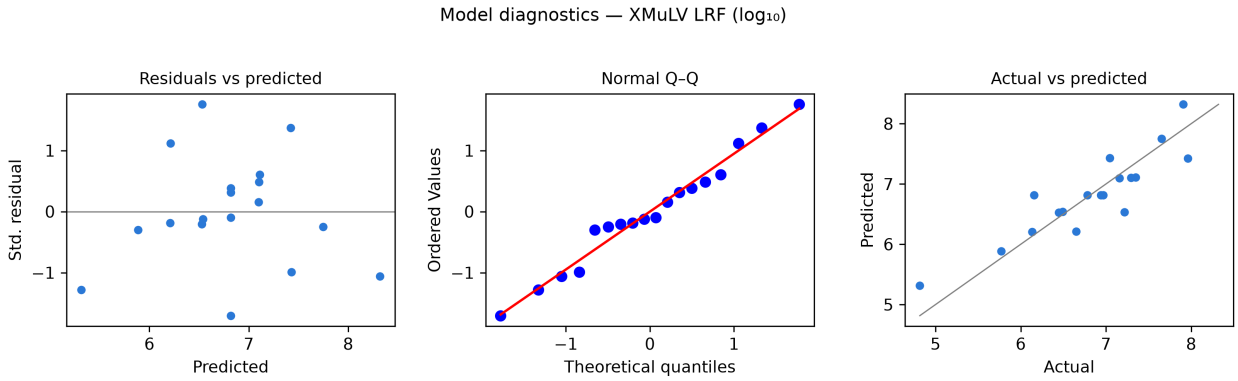


Figure 5.2: Diagnostics of the selected model of the XMuLV log reduction. Left, standardized residuals against predicted values. Middle, normal probability plot of the residuals. Right, predicted against measured values.

Figure 5.3 shows the residuals of the two selected models against the order of execution, and the Cook's distance of each run against the cut-off.

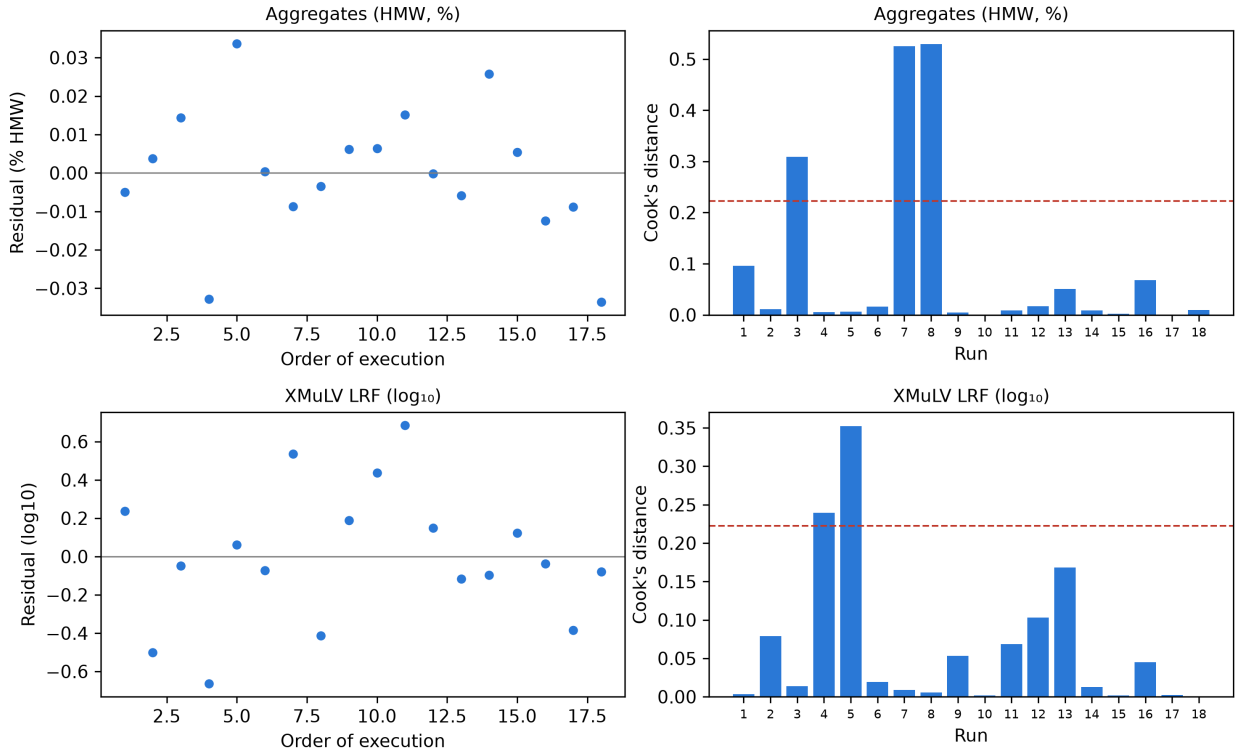


Figure 5.3: Residuals of the selected models against the order of execution (left) and the Cook's distance of each run against the cut-off of 4/n (right).

Both selected models meet the criterion on residuals and influential runs, the fourth of PCMP-001 §7.2. Neither regression of that test is significant for either model. For aggregate the absolute residuals do not grow with the predicted value ($p = 0.194$)

and the residuals do not drift with the order of execution (p = 0.426), and the fitted drift over the 18 runs is −0.011 percentage points, against a margin of 4.92 points between the highest single-batch prediction over the NOR and the criterion. For the XMuLV log reduction the two p-values are p = 0.989 and p = 0.758, and the fitted drift is 0.09 $\log_{10}$ against a margin of 0.17 $\log_{10}$ between the lowest mean prediction over the NOR and the claim. That drift is 52 % of the margin, and it is not significant.

No run dominates either fit. The criterion fails where a run's Cook's distance exceeds the 50 % quantile of the F distribution with the degrees of freedom of the model, which is 0.83 for the aggregate model and 0.88 for the XMuLV model. The largest Cook's distance is 0.53 at run 8 for aggregate and 0.35 at run 5 for the XMuLV log reduction, so neither model fails. Both lie above the cut-off of 0.22 at which PCP-006 §7.1 calls for an examination. Three runs of the aggregate model are above that cut-off, runs 3, 7 and 8, and two of the XMuLV model, runs 4 and 5. Section 5.3.2 gives the examination of each.

5.3.1 Models near the selected one

For three outputs another model lies within 2 AICc units of the selected one. Table 5.9 lists those models with their adequacy and the judgement each gives. The comparison is registered within PD-006-04.

Table 5.9: Models within 2 AICc units of the selected model, and the judgement each gives over the NOR in force.

Output	ΔAICc	Terms	R² minus predicted R²	All four criteria	Judgement over the NOR	Met
Aggregates (HMW, %)	0.00	hold time and temperature	0.004	Met	upper limit 2.10 %	Yes
	1.66	hold time, temperature and temperature²	0.005	Met	upper limit 2.10 %	Yes
	1.75	hold time, temperature and hold time × temperature	0.007	Not met, influential run	upper limit 2.10 %	Yes
XMuLV LRF (log_{10})	0.00	inactivation pH, hold time and temperature	0.120	Met	lowest mean 6.77 log10	Yes
	0.35	inactivation pH, hold time, temperature and inactivation pH²	0.124	Met	lowest mean 6.93 log10	Yes
	1.91	inactivation pH and hold time	0.147	Met	lowest mean 6.77 log10	Yes
Fragments (LMW, %, nrCE-SDS)	0.00	inactivation pH, hold time, inactivation pH² and hold time²	0.328	Not met, overfitting	upper limit 1.11 %	n.a.
	0.77	inactivation pH, hold time, temperature, hold time × temperature, inactivation pH² and hold time²	0.273	Not met, overfitting	upper limit 1.09 %	n.a.
	1.38	hold time	0.169	Met	upper limit 1.12 %	Yes
	1.57	inactivation pH, hold time and inactivation pH²	0.345	Not met, overfitting	upper limit 1.14 %	n.a.

ΔAICc, the AICc of the model minus the AICc of the selected model, which is the first row of each output. All four criteria of PCMP-001 §7.2, with the reason where one is not met. Overfitting, R² minus predicted R² above the limit. Influential run, a Cook's distance above the F limit, or a significant residual regression. Judgement over the NOR: for the XMuLV log reduction the lowest mean prediction among the NOR combinations against the claim of the step, for aggregate and fragments the highest upper end of the single-run prediction interval against the criterion of the attribute, aggregate at manufacturing scale. Where a model is not adequate the judgement is shown for information and no range follows from it.

For aggregate the three models within 2 units share hold time and temperature and differ in a squared temperature term and a hold time by temperature interaction. All three give an upper single-batch limit within 0.003 percentage points of one another, and all three meet the criterion across the NOR. The model with the interaction does not meet all four adequacy criteria, because run 3 carries a Cook's distance above the F limit in it, and the selected model decides in any case. For the XMuLV log reduction the three models share pH and hold time, two of them keep temperature and one adds a squared pH term. All three meet the four criteria. The lowest mean prediction over the NOR ranges from 6.77 to 6.93 $\log_{10}$ among them, and the claim of 6.60 $\log_{10}$ is met under each. For fragments the four models within 2 units disagree about the form of the surface. Three of them carry a squared pH term and exceed the limit on overfitting, and one of those three also fails the criterion on residuals and influential runs. The fourth carries hold time alone and meets all four criteria, and it lies 1.38

AICc units above the selected model. The selected model decides, so fragments set no range, and their predicted levels are not carried into any conclusion of this report.

5.3.2 Influence analysis

Table 5.10 gives the examination PCP-006 §7.1 requires of each flagged run, by the procedure of Section 3.5. Every flagged run lies inside the range studied, and none was excluded from the analysis the rest of this report uses.

Table 5.10: Each run above the Cook's distance cut-off deleted in turn, with the model then selected and the judgements it gives.

Output	Run deleted	Cook's distance	Terms of the model then selected	First three criteria	Judgement over the NOR in force	PARs
Aggregates (HMW, %)	3	0.31	hold time, temperature and hold time × temperature	Met	2.09 %, met	unchanged
	7	0.53	hold time, temperature and temperature^2	Met	2.09 %, met	unchanged
	8	0.53	hold time and temperature	Met	2.09 %, met	unchanged
XMuLV LRF ($\log_{10}$)	4	0.24	inactivation pH, hold time and temperature	Met	6.80 log10, met	wider
	5	0.35	inactivation pH and hold time	Met	6.84 log10, met	wider

Cook's distance of the run in the selected model of all runs, against the cut-off of 4/n. First three criteria of PCMP-001 §7.2: the model F-test, lack of fit and the limit on overfitting. Judgement over the NOR in force: for the XMuLV log reduction the lowest mean prediction among the NOR combinations against its claim, for aggregate the highest upper end of the single-run prediction interval at manufacturing scale against its criterion. PARs: whether the proven acceptable range of each factor is the same as under the selected model of all runs. The text gives the wider ranges.

Four conclusions of this report hold under every deletion. The NOR in force stays inside the PAR of each of the three factors, the claim is met across the NOR in force, the pH NOR still has to move to meet it, and inactivation pH stays a critical process parameter, hold time and temperature well-controlled critical process parameters and A-Mab concentration a general process parameter. For aggregate the model selected after a deletion keeps hold time and temperature in every case and adds a third term in two of the three. Temperature raises aggregate by 0.204 to 0.226 percentage points over the range studied in those models, against 0.215 points on all runs, and the effect is significant in every one of them ($p < 0.001$ at its weakest). The upper single-batch limit over the NOR moves by at most 0.014 percentage points, the criterion is met in every case, and the highest single-batch limit anywhere in the region studied is 2.39 %, still below 7.02 %. The one-sided 95 % prediction bound of the aggregate increase within the NOR as characterized, which replaced the platform worst case upstream of this step, is at most 0.55 percentage points after any one deletion, against 0.56 on all runs and a platform value of 0.70, so it stays below the platform value in every case (Section 7.7).

Two results of the XMuLV model do depend on run 5. With that run deleted the temperature term leaves the model, so the temperature effect on the log reduction of 0.57 $\log_{10}$ (p = 0.036) rests on it. The PAR of each factor widens when the run is deleted. The pH PAR runs to pH 3.68 instead of pH 3.61, the hold-time PAR begins at 72 min instead of 84 min, and the temperature PAR covers the whole range studied. The deletion of run 4 keeps all three terms and moves the PARs little. Under every model the ranges reported in Section 7.4 are the narrowest of the three, so the NOR in force lies inside the PAR under each of them, and the lowest mean prediction over the NOR in force is 6.80 and 6.84 $\log_{10}$, above the claim in both cases. Over the NOR as characterized the claim fails under both deleted models as it does under the selected model, at 6.12 and 6.38 $\log_{10}$ against 6.16 $\log_{10}$, so the decision to move the NOR does not rest on one run.

5.4 Mechanistic interpretation

Figure 5.4 shows the predicted XMuLV log reduction over pH and hold time at the two ends of the temperature NOR in force.

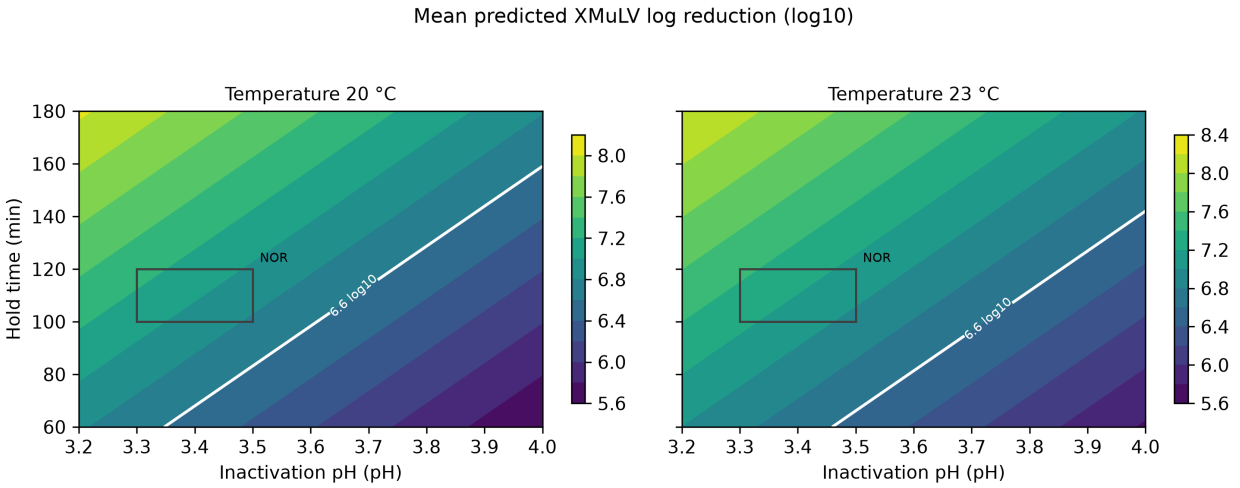


Figure 5.4: Predicted XMuLV log reduction over inactivation pH and hold time, at the two ends of the temperature NOR in force. The claim of the step is drawn as a white contour, the NOR in force as a grey box. Lighter shading is a larger log reduction.

The log reduction falls as the pH rises and grows as the hold lengthens, and the contour of the claim runs from the bottom left of each panel to the top right. The surface the data give is a plane in the three factors. It carries no interaction of pH with hold time, so the hold time that restores the claim after a rise in pH is the same at every pH within the range studied, and it carries no curvature, so the log reduction grows at a constant rate with the hold time. The first-order description of Section 2 predicts the opposite on both counts. A log reduction of k(pH, T) multiplied by the hold time rises faster with time the lower the pH, which is an interaction, and these

data show none. On the full quadratic model of the planned runs that interaction lies between −0.27 and 0.92 $\log_{10}$ over the corners of the range studied (p = 0.246), so 18 runs bound it rather than resolve it. The first-order description does not explain the plane, and the surface reported here is what the data give over the range studied. The three factors trade against one another on it. A rise of 0.10 pH units, the half-width of the pH NOR, lowers the log reduction by 0.15 $\log_{10}$, which a hold longer by 15 min or warmer by 2.7 °C offsets. A pool at manufacturing scale reaches its pH by acid addition and mixing, and a pH above the one intended lowers the log reduction. That is why the claim is most exposed to the inactivation pH (Section 9).

Aggregate grows with the hold time and with the temperature, which is the direction the association of partially unfolded antibody gives (Section 2). No interaction of hold time and temperature was resolved, and one of the models within 2 AICc units carries one (Section 5.3.1), so the data do not decide whether the two effects add or interact. Within the NOR as characterized the predicted rise across the step is 0.53 percentage points, with a single-batch bound of 0.56 points (Section 7.7). The mechanism also leads one to expect faster aggregation at lower pH, because a more acid pool is further unfolded. No such effect was resolved within the range studied, and the data bound a linear effect within about 0.03 points over that range (Section 5.3), so within this range the hold time and the temperature determine the aggregate the step adds.

The acidic variants grow with the hold time and with nothing else, by 0.20 percentage points per 60 min. No effect of pH and no effect of temperature was resolved on them in any study of the step. The acid-catalysed hydrolysis of side-chain amides that Section 2 describes would speed up at a lower pH and at a higher temperature, so it does not explain results that depend on the hold time alone. The rise over the whole hold-time range studied, 0.4 points, is small against the level of about 21 % in the neutralized pool, which is formed upstream, and the attribute stays far inside its criterion at every setting of the step (Section 7.2).

Fragments grow with the hold time as well. S-VI-OFAT resolved an increase of 0.048 percentage points per 60 min at the two pH values it compared (Section 7.5.1), in the direction cleavage in the hinge region leads one to expect. At that rate a hold at the upper end of the NOR in force adds about 0.10 points. The attribute leaves 1.47 points to its criterion (Section 7.2). No effect of pH and no effect of temperature on fragments was resolved in any study.

6 Design space

The design space of the step is the region of inactivation pH, hold time and temperature over which every attribute with a criterion at this step meets it. PCP-006 §7.6 states how it is drawn. It was evaluated on a grid of 11 levels of each factor over the ranges studied, 1,331 points in all, with each output judged on the statistic Section 3.5 defines for it. Aggregate is inside when both ends of the 95 % prediction interval of a single run lie within its criterion at manufacturing scale, and the XMuLV log reduction is inside when its mean prediction reaches the claim.

Of the region studied, 63.9 % meets the criteria of the two outputs that have an adequate model. XMuLV clearance alone rejects 36.1 % of the grid, and aggregate rejects none of it. The region is therefore bounded by the clearance claim and not by product quality, and its boundary is the contour drawn in Figure 5.4. Judging both outputs on their mean predictions gives the same region, because aggregate stays far from its criterion at every point of the grid, so the choice of statistic does not move this boundary. Acidic variants and fragments have no adequate model, so neither draws a boundary. Every measured result of both attributes meets its criterion across the designs, which span the whole region studied, and the region is not constrained by either attribute on that evidence.

The NOR in force lies inside the design space, with the margins Section 7.5 gives, and the region studied is the knowledge space of this report. Movement within the design space is not a change from a regulatory filing perspective, as ICH Q8(R2) states, and is assessed under the quality system of ICH Q10 (International Council for Harmonisation 2008). A move outside the NOR but inside the design space needs a prospective risk assessment under that system. A move outside the region studied has no evidence in this report and needs new characterization data.

Two limitations apply to the region. The models were fitted in the scale-down model and the spiking model of the step, and the clearance boundary rests on the spiking model, whose representativeness is argued in Section 3.1 rather than measured at scale. The boundary is drawn on the mean predicted log reduction, so at the boundary itself a single batch falls below the claim about half the time, and the step is operated at the set-point with the margin Section 7.5 states.

7 Proven acceptable ranges

7.1 The acceptance basis

A drug substance criterion does not say what the output of an upstream step must meet. Each output of this step is therefore judged against an intermediate acceptance criterion, which PCMP-001 §7.3 derives by carrying the drug substance criterion backward through the train with the worst case that each downstream step showed on the platform, the lowest clearance factor for a step that removes the attribute and the largest increase for a step that adds to it. No margin is applied. An attribute that no downstream step changes keeps its drug substance criterion. Table 7.1 gives the four criteria of this step and their derivation.

Table 7.1: Intermediate acceptance criteria for the low-pH viral inactivation output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Aggregates, HMW (%)	≤ 5.00	1.40	0.00	≤ 7.02	Carried back, tightened[a)]
Fragments, LMW (%)	≤ 2.00	1.30	0.00	≤ 2.60	Carried back
Acidic variants (%)	18.0–40.0	1.00	0.00	18.0–40.0	DS criterion[b)]
XMuLV clearance (log10)	≥ 16.7[c)]	not applicable		≥ 6.60	Platform claim[d)]

[a)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[b)] No downstream step changes the attribute, so the drug-substance criterion applies.
[c)] Cumulative requirement across the steps that clear the virus.
[d)] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

Aggregate in the neutralized pool must not exceed 7.02 % HMW. Cation exchange is the only step after this one that is credited with removing aggregate, so the criterion is the drug substance criterion of 5.00 % multiplied by the clearance factor of cation exchange. Within its NOR, cation exchange cleared less aggregate than

its platform worst case, so the one-sided 95 % prediction bound of a single cation exchange batch, a clearance factor of 1.40, replaced the platform value, and the a priori criterion of 10.0 % tightened to the value in Table 7.1. That analysis is in PCR-007, the characterization report of cation exchange. Fragments must not exceed 2.60 % by nrCE-SDS, which is the drug substance criterion of 2.00 % carried back through the platform worst case of cation exchange, 1.30-fold. Acidic variants must lie within 18.0 to 40.0 %, the drug substance criterion, because no downstream step is credited with a change in them, and both ends apply. XMuLV clearance must reach the platform claim of the step, 6.60 $\log_{10}$.

The criteria were derived after the studies ran, in PCMP-001 Version 2.0. That is acceptable because the derivation uses no result of the studies it judges. The drug substance criteria come from the CQA register, and the downstream worst cases come from the platform history of X-Mab, Y-Mab and Z-Mab. A-Mab data enter a criterion only where verification replaced a platform worst case with the prediction bound of a single batch of the same downstream step, which is what happened to the aggregate criterion of this step. Table 1.2 compares them with the limits of Version 1.0.

The cumulative requirement for XMuLV is met by three claims. Low-pH inactivation claims 6.60 $\log_{10}$, anion exchange 5.50 $\log_{10}$ and virus filtration 4.62 $\log_{10}$, which sum to 16.72 $\log_{10}$ against a requirement of 16.7 $\log_{10}$. The surplus is 0.02 $\log_{10}$, so a shortfall at this step larger than that cannot be made up by the other two claims. PCMR-001 consolidates the cumulative claim.

7.2 The outputs against their criteria

Table 7.2 gives each output against its criterion across the NOR in force.

Table 7.2: Low-pH viral inactivation outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model, or measured results where it has none[a)b)]		95 % limits of a single run over the NOR[b)]	Met across the NOR[c)]
		At the set-point	Range over the NOR		
Aggregates, HMW (%)	≤ 7.02	1.98	1.92 to 2.06	1.88 to 2.10	Yes
XMuLV clearance (log10)	≥ 6.60	7.08	6.77 to 7.45	n.a.[d)]	Yes
Acidic variants (%)	18.0–40.0	n.a.[a)]	20.2 to 20.6[a)]	n.a.[a)]	Yes
Fragments, LMW (%)	≤ 2.60	n.a.[a)]	0.87 to 1.13[a)]	n.a.[a)]	Yes

[a)] No adequate model (PCMP-001 §7.2): the range of the measured results of the designs, less the offset of the scale-down model where one is known, on which the output is judged.
[b)] At manufacturing scale: the scale-down prediction minus the offset of the scale-down model for this output.
[c)] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty). Viral clearance: the mean prediction.
[d)] Judged on the mean prediction at the least favourable combination, the lower end of the range over the NOR.

Every output meets its criterion. The fourth column of Table 7.2 holds the statistic that decides each quality attribute, the 95 % limits of a single run over the NOR. For aggregate those limits are 1.88 to 2.10 % HMW at manufacturing scale, against a criterion of 7.02 %, which leaves 4.92 percentage points. The upper limit falls at pH 3.3, 120 min and 23 °C, the longest hold and the warmest temperature of the NOR. At the set-point the model predicts 1.98 % HMW. The conclusion holds with the offset of the scale-down model at either end of its interval, as Section 3.1 reports.

The XMuLV log reduction is predicted at 7.08 $\log_{10}$ at the set-point. The lowest mean prediction over the NOR in force is 6.77 $\log_{10}$, at pH 3.5, 100 min and 20 °C, the highest pH, the shortest hold and the coolest temperature of the NOR. The claim of 6.60 $\log_{10}$ is met there, with 0.17 $\log_{10}$ to spare. The prediction interval of a single batch at that combination runs from 5.90 to 7.64 $\log_{10}$, so a single spiked run at the least favourable corner can fall below the claim. That interval is reported for information (Section 3.5). Of the 29 individual results of the designs, 8 lie below the claim. Every run of both designs was made at a corner, a face centre or the centre of the region studied, so none was made inside the NOR in force, and the lowest of those results was measured at pH 4, 60 min and 15 °C, outside the NOR in every factor.

Acidic variants and fragments have no adequate model, so each is judged on its measured results, as PCMP-001 §7.2 requires. The acidic variants of the designs run from 20.2 to 20.6 % at manufacturing scale, inside the criterion of 18.0 to 40.0 % at both ends and far from either. The conclusion holds with the offset at either end of its interval. The fragments of the response-surface design run from 0.87 to 1.13 %, against a criterion of 2.60 %. The designs span the whole region studied, which contains

the NOR, so these results cover the NOR as well. They are stated at the scale of the laboratory model for fragments, whose offset was not measured (Section 3.1).

7.3 Hold of the neutralized pool

PCP-006 planned no hold study, and Version 1.0 of this report had no hold result. The neutralized pool was therefore held during the Version 2.0 re-analysis, from 2026-09-20 to 2026-09-29, registered as PD-006-05, the hold study of the neutralized pool. One pool made at the set-points as characterized was held at 21 °C and 25 °C for 7 days, and aggregate, acidic variants and fragments were measured at every time point. Table 7.3 gives the results.

Table 7.3: Hold-time stability of the neutralized viral inactivation pool.

Storage	Day	Size variants (%)		Acidic variants (%)
		HMW (SEC)	LMW (nrCE-SDS)	
21 °C	0	1.68	0.93	22.8
	1	1.74	1.10	22.6
	3	1.84	1.07	22.5
	5	2.03	1.00	22.4
	7	2.06	1.18	22.8
25 °C	0	1.75	0.94	22.2
	1	1.76	1.04	22.5
	3	1.95	1.14	22.8
	5	2.05	1.10	23.4
	7	2.09	1.10	23.8

PCP-006 sets no hold criterion. A held pool must meet the criteria of Table 7.1 when it enters cation exchange, as PCMP-001 §7.3 requires, so what has to be judged is what the hold adds to the level the step delivers. The approach is the following. The highest level expected to enter the hold is taken from the step itself, which is the upper single-batch limit over the NOR for aggregate and the highest measured result for the two outputs with no model, and the largest increase measured over the hold is added to it. No plan sets that approach, and it was chosen after the hold results were known, which PD-006-05 records (Section 13).

Aggregate rose by at most 0.383 percentage points over 7 days, more than the 0.072 points between the two aliquots at day 0, so the increase is resolved. Added to the 2.10 % expected to enter the hold it gives 2.48 %, against the criterion of 7.02 %. Fragments rose by at most 0.249 points, against 0.008 points between the day-0 aliquots, which gives 1.38 % against the criterion of 2.60 %. The acidic variants rose by 1.58 points at 25 °C, which gives 22.2 % against the upper limit of 40.0 %, and the lowest level expected to enter the hold is 20.2 % against the lower limit of 18.0 %. At 21 °C no change in acidic variants was resolved against the 0.59 points between the day-0 aliquots. The upper limits for aggregate and fragments are met, and so are both ends of the acidic variant criterion.

The pool itself is also to be read as it was measured. Table 7.3 gives 1.68 to 1.75 % HMW and 22.2 to 22.8 % acidic variants at day 0, and 23.8 % acidic variants at day 7 at 25 °C, the highest level of the study. At manufacturing scale those acidic variant values are 21.9 to 22.5 % at day 0 and 23.5 % at the end, inside the criterion at both ends. The day-0 values lie above every acidic variant result of S-VI-OFAT, which ran in the same days and reached 21.1 % at most at manufacturing scale (Section 7.5.1). The record gives no reason for the difference. The conclusion covers that one pool, at the two temperatures and over the time tested. The study used material made at the set-points as characterized, so it bounds what a hold adds to a pool of that composition and not what it adds to a pool from the edge of the NOR.

7.4 The ranges of the three factors

Table 7.4 gives the PAR of each factor for XMuLV clearance, as PCMP-001 §6.4 defines the two analyses. The first analysis scans the factor across its range studied with the other two at their set-points and compares the mean prediction with the criterion. The second varies the other two within their NORs, as Section 3.5 sets out, and compares the central 95 % of the simulated batches with the criterion for a quality attribute and the lowest mean prediction among the NOR combinations with the claim for viral clearance. The second analysis is the one that supports a claim, because the range must hold while the rest of the operating point moves within its NOR.

Table 7.4: Proven acceptable ranges of the three factors for XMuLV clearance, in natural units, from the selected response-surface model.

Parameter	Range studied	Criterion	PAR (set-point)	PAR (NOR)
Inactivation pH (pH)	3.2–4	≥ 6.60	3.2–3.74	3.2–3.61
Hold time (min)	60–180	≥ 6.60	98.4–180	84–180
Temperature (°C)	15–25	≥ 6.60	16.2–25	17.2–25

Range studied, the characterized range. Criterion, the claim of the step in log10. PAR (set-point), the first analysis of PCMP-001 §6.4. PAR (NOR), the NOR-propagated analysis, which is the one that supports the claim. A range that reaches the end of the range studied is reported as reaching it, and no claim is made beyond it. Aggregate, acidic variants and fragments are not shown. The text gives their ranges.

One edge of each range comes from XMuLV clearance. Its PAR is pH 3.2–3.61, 84–180 min and 17.2–25 °C. The upper pH edge and the lower hold-time and temperature edges are where the mean predicted log reduction falls to the claim. The other three edges are the ends of the ranges studied, so they are where the study stopped and not where the criterion fails. Aggregate meets its criterion across the whole range studied of each factor on the selected model, so its PAR is the range studied in all three. Acidic variants and fragments have no adequate model, so neither sets a PAR. Their measured results meet their criteria throughout the designs, which span the region studied (Section 7.2). The range carried into the control strategy is therefore

the XMuLV range of each factor. Those ranges widen if run 5 is deleted from the design (Section 5.3.2), and the ranges reported here are the narrower ones.

Figure 7.1 and Figure 7.2 show the two analyses for one parameter of each modelled attribute, inactivation pH for the XMuLV log reduction and hold time for aggregate. Each is the parameter with the largest coefficient in its model, although for XMuLV clearance the coefficients of pH and hold time are of the same size (Section 5.3).

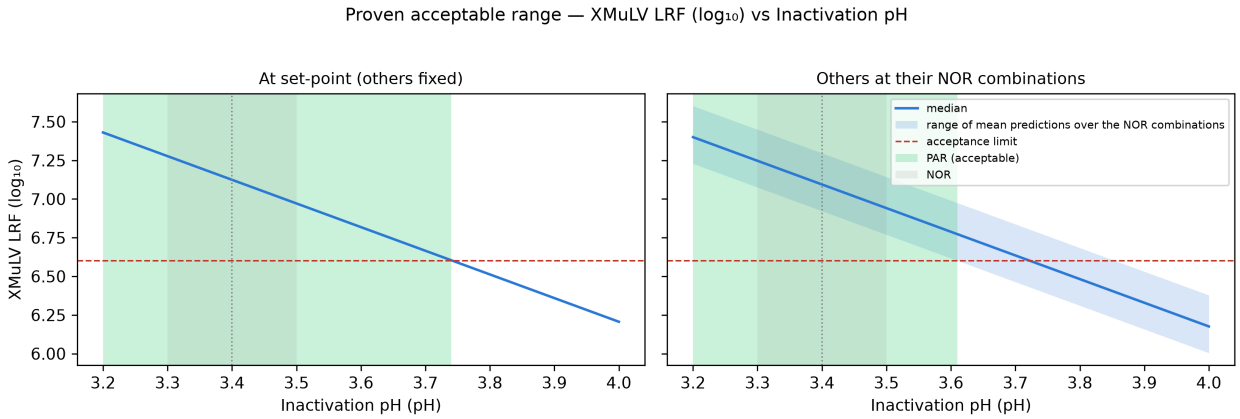


Figure 7.1: Proven acceptable range of the inactivation pH for the XMuLV log reduction. Left, the mean prediction with the other two factors fixed. Right, the median and the range of mean predictions over the NOR combinations of hold time and temperature. In both panels the band labelled PAR in the legend is the acceptable range of the pH, the band labelled NOR is the NOR in force, the dotted vertical line is the set-point and the dashed horizontal line is the claim of the step.

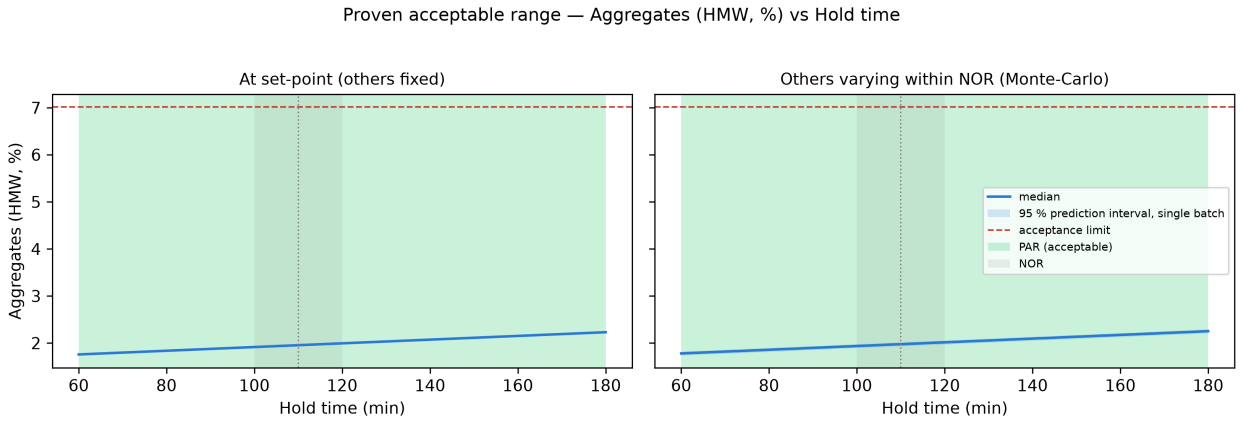


Figure 7.2: Proven acceptable range of the hold time for aggregate, at manufacturing scale. Left, the mean prediction with the other two factors fixed. Right, the median and the 95 % prediction interval of a single batch with pH and temperature varying within their NORs. The band labelled PAR in the legend is the acceptable range, the band labelled NOR is the NOR in force and the dotted vertical line is the set-point. The dashed horizontal line is the criterion of the attribute, and the axis reaches it, so the predicted rise along the hold time is a shallow slope at the foot of the panel.

In the left panel of Figure 7.1 the mean predicted log reduction falls through the claim inside the range studied, and in the right panel the band is the range of mean predictions over the NOR combinations of hold time and temperature. In Figure 7.2 the predicted aggregate rises from 1.78 % at 60 min to 2.25 % at 180 min with the other two factors at their set-points, a rise of 0.47 percentage points, against a criterion of 7.02 %. The axis of the panel reaches the criterion, so that rise appears as a shallow slope and the shaded acceptable region covers the whole panel.

The ranges above and the design space of Section 6 are two views of the same models. A PAR is one factor scanned with the others inside their NORs, and the design space is the region in all three factors at once. The control strategy uses the NOR, which lies inside both.

7.5 The set-points and ranges in force, and the decisions behind them

The studies were planned against the set-points and NORs of PCP-006 Version 1.0 and run around them. Against those values the XMuLV claim was not met. The lowest mean predicted log reduction over the NOR as characterized was 6.16 $\log_{10}$, at pH 3.6, 60 min and 19 °C, against the claim of 6.60 $\log_{10}$. At the former pH set-point of pH 3.5 the lowest mean prediction over that NOR is already 6.31 $\log_{10}$, so no PAR of any factor contained that NOR, and PCP-006 §8 rule 4 requires that to be reported as no PAR. Table 7.5 gives the values the studies were run around and the values in force.

Table 7.5: Set-points and normal operating ranges as characterized and in force.

Parameter	Unit	Set-point as characterized	NOR as characterized	Set-point in force	NOR in force	Change
Inactivation pH	pH	3.5	3.4–3.6	3.4	3.3–3.5	set-point moved and NOR moved
Hold time	min	90	60–120	110	100–120	set-point moved and NOR narrowed
Temperature	°C	21	19–23	21	20–23	set-point unchanged and NOR narrowed

As characterized, the value the studies were planned against and run around. In force, the value this revision adopts. The A-Mab concentration is unchanged, and its NOR equals the range studied.

Three decisions produced the values in force, each taken after the results of the designs were known. The pH at which the claim is lost depends on the shortest hold and the coolest temperature of the NOR, because a longer and warmer hold raises the log reduction. The two decisions that narrowed the hold-time NOR therefore moved that pH, and the decision that re-centred the pH left it where the first had put it.

1. The hold-time NOR was narrowed to begin at the former set-point of 90 min, giving 90–120 min, and the temperature NOR to 20–23 °C. With those ranges the model puts the pH at which the claim is lost at 3.54. The pH NOR as characterized reached 0.06 pH units beyond it, and the former set-point of pH 3.5 sat 0.04 pH units inside it.
2. The pH set-point was moved to pH 3.4 and the pH NOR to pH 3.3–3.5, and the hold-time set-point to 105 min, the middle of the hold-time NOR of the first decision. A NOR of −0.10 to +0.10 pH units about the set-point is already tight for pH control at manufacturing scale, so the width of the NOR was kept and the range was placed at the acid end of the NOR as characterized, where the log reduction is larger. The upper pH NOR edge of pH 3.5 then had a margin of 0.04 pH units to the pH at which the claim is lost. Two studies support the new pH set-point, a hold of the pool at it against the former set-point (Section 7.5.1) and cation exchange runs on material inactivated at it (Section 7.5.2).
3. A margin of 0.04 pH units is smaller than the 0.10 pH units the NOR itself allows on either side of the set-point, which is about what pH control holds at manufacturing scale. No plan sets a minimum margin, so the size that is enough is a judgement of this report. On that judgement 0.04 pH units was not accepted, and the hold-time NOR was narrowed again, to 100–120 min, with its set-point at the middle of it, 110 min. With the ranges in force the model puts the pH at which the claim is lost at 3.61, a margin of 0.11 pH units from the upper pH NOR edge, which was accepted. A narrower hold-time NOR moves routine operation toward longer holds, which add aggregate, fragments and acidic variants, so confirmation runs were made at the low end of the pH PAR with hold time and

temperature at the upper ends of their NORs (Section 7.5.3).

Each pH is computed from the selected model at the shortest hold and the coolest temperature of the NOR of that stage. Table 7.6 gives the margins of the NOR-propagated analysis of Section 7.4.

Table 7.6: Margins from the NOR in force to the proven acceptable range set by XMuLV clearance.

Factor	Unit	NOR in force	PAR	Edge of the PAR that binds	Margin to NOR edge	Margin to set-point
Inactivation pH	pH	3.3–3.5	3.2–3.61	upper	0.11	0.21
Hold time	min	100–120	84–180	lower	16	26
Temperature	°C	20–23	17.2–25	lower	2.8	3.8

PAR, the NOR-propagated proven acceptable range of the factor for XMuLV clearance. Margins in the unit of the factor, computed from the PAR edges as printed. The other edge of each range is the end of the range studied.

The NOR in force lies inside the PAR of every attribute, so the step meets every criterion across it (Section 7.2). The edge of the pH NOR that binds carries a margin of 0.11 pH units and the set-point 0.21 pH units, both above the 0.10 pH units of the judgement above.

These changes are changes to the control strategy of the step. They were decided on the data of the studies reported here, which means the ranges in force were chosen after the results were seen, and the confirmation runs of Section 7.5.3 were made for that reason. The comparisons with the platform worst cases (Section 7.7) and with Version 1.0 use the NOR and set-points as characterized, which the studies were planned against, so that no change of a NOR moves a criterion or a Version 1.0 number.

7.5.1 Quality attributes at the new pH set-point

PCP-006 planned no study at the Version 2.0 set-points. The response-surface design covers pH 3.2 to 4, so it predicts the attributes at the new set-point rather than measuring them there. Study S-VI-OFAT therefore held the pool at pH 3.4 against pH 3.5 as a control, at 21 °C, over hold times of 0, 60, 90, 105, 120 and 180 min, in duplicate at each pH and hold time, 12 runs per pH. It ran from 2026-09-21 to 2026-09-23 and is registered as PD-006-01, the study at the new pH set-point. Table 7.7 gives every result and Table 7.8 the analysis.

Table 7.7: Quality attributes over the low-pH hold at the re-centred and the former pH set-point (study S-VI-OFAT).

Hold time (min)	Replicate	Aggregate, % HMW (SEC)		Fragments, % LMW (nrCE-SDS)		Acidic variants, % (icIEF)	
		pH 3.40	pH 3.50	pH 3.40	pH 3.50	pH 3.40	pH 3.50
0	1	1.65	1.62	0.94	1.03	19.99	20.76
	2	1.50	1.57	0.92	0.81	20.29	20.27
60	1	1.73	1.80	1.04	0.92	20.50	20.88
	2	1.76	1.76	0.80	0.98	20.07	20.20
90	1	2.07	1.83	0.97	1.11	20.59	20.81
	2	1.87	1.91	1.01	0.81	20.50	20.98
105	1	1.95	2.09	1.08	0.98	20.44	20.57
	2	2.13	1.92	1.03	1.00	19.83	20.95
120	1	1.99	2.07	1.03	1.06	20.01	20.89
	2	2.14	2.00	0.94	1.00	20.64	20.30
180	1	2.14	2.46	1.06	1.00	21.19	21.39
	2	2.35	2.29	1.02	1.12	20.32	20.93

Temperature 21 °C. Time 0 is the load after pH adjustment, before the hold. Each hold time was run in duplicate at each pH.

Table 7.8: Effect of the inactivation pH on the quality attributes over the low-pH hold (study S-VI-OFAT).

Attribute	Method	Mean		Effect of pH 3.40 against pH 3.50		pH × hold time, p[a)]	Change per 60 min[b)]	Method repeat-ability, SD[c)]	Results at pH 3.40[d)]	Criterion of the step
		pH 3.40	pH 3.50	Difference (95 % CI)[e)]	p[f)]					
Aggregate	SEC	1.94	1.94	0.00 (−0.08 to +0.07)	1.000	0.568	+0.25	0.07	1.50–2.14	≤ 7.02
Fragments	nrCE-SDS	0.99	0.98	0.00 (−0.06 to +0.07)	1.000	0.828	+0.05	0.06	0.80–1.08	≤ 2.60
Acidic variants	icIEF	20.36	20.74	−0.38 (−0.65 to −0.11)	0.025	0.875	+0.19	0.41	19.83–20.64	18.0–40.0

[a)] Interaction: whether the change with hold time differs between the two pH values.
[b)] Common to both pH values, in the unit of the attribute.
[c)] At the mean level of the attribute in the study.
[d)] Lowest and highest single result at hold times up to 120 min, the upper end of the hold time NOR in force.
[e)] Difference of the means at any hold time, from a model of the result on pH (two levels) and hold time, in the unit of the attribute (points for a percentage).
[f)] Adjusted by Holm’s method for the three attributes tested.

Each attribute was analysed on the pH, as a two-level factor, and the hold time, so the difference between the two pH values is estimated at any hold time. The interaction was tested separately, and the three pH p-values were adjusted by Holm's method, because three attributes were tested.

Aggregate and fragments rose with hold time, and for neither was an effect of pH detected. Aggregate rose 0.248 percentage points per 60 min ($p < 0.001$) and fragments 0.048 points per 60 min ($p = 0.012$). The difference between the two pH values was −0.004 points for aggregate, with a 95 % confidence interval from −0.079 to 0.072 points, and 0.003 points for fragments, from −0.063 to 0.068 points. Neither reached significance after adjustment, and each interval is narrow against the margin of the attribute to its criterion.

The acidic variants were 0.38 percentage points lower at pH 3.4 than at pH 3.5, with a 95 % confidence interval from 0.11 to 0.65 points, and the difference is significant after adjustment ($p = 0.025$). Three features place it before the hold rather than in it. The difference was already 0.37 points at time 0, in the pH-adjusted load before any hold, where a hold effect cannot yet have acted. It did not grow with hold time ($p = 0.875$), and a reaction driven by the hold pH would. The response-surface model of the attribute carries hold time alone and no pH term (Section 5.3). The difference is 0.38 percentage points against a criterion 22.0 points wide, and the lowest result at the new set-point, 19.6 % at manufacturing scale, lies 1.6 points above the lower limit of 18.0 %. No plan sets a size at which a significant difference of this attribute would matter, so calling this one small against its criterion is a judgement of this report.

Every result at the new pH set-point meets the criterion of its attribute. Over the hold times of the study up to the upper end of the hold-time NOR in force, which are 0, 60, 90, 105 and 120 min, the results run from 1.48 to 2.12 % HMW for aggregate at manufacturing scale, against 7.02 %, from 0.80 to 1.08 % for fragments, against 2.60 %, and from 19.6 to 20.4 % for acidic variants at manufacturing scale, against 18.0 to 40.0 %. At the two hold times inside the NOR in force, 105 and 120 min, aggregate runs from 1.93 to 2.12 %, fragments from 0.94 to 1.08 % and acidic variants from 19.6 to 20.4 %. Each conclusion for aggregate and acidic variants holds with the offset at either end of its interval. The study measured no viral clearance, so it supports the new set-point for product quality only, and the clearance there comes from the model (Section 7.2). Fragments were measured by nrCE-SDS in this study, as Table 3.3 shows.

7.5.2 Cation exchange on material from the step at the new pH set-point

The criteria of this step assume that cation exchange clears aggregate and fragments from its load by at least a fixed factor (Section 7.1). PCP-006 planned no downstream verification. Study S-CEX-VER therefore ran cation exchange at set-point on material from the step at its new pH set-point, inactivated at pH 3.4 for 120 min at 23 °C, the longest hold and the warmest temperature of the NOR in force. It ran from 2026-09-24 to 2026-09-25 and is registered as PD-006-02, the cation exchange verification runs.

The number of runs was fixed before the runs were made, by the simulation in Table 7.9. The clearance factor of a run is the load level divided by the pool level, and its logarithm is the log reduction value of that run. A verification passes when the one-sided 95 % lower confidence bound of the mean log reduction value of the runs reaches the clearance factor the criteria rely on. The simulation gives the smallest number of runs, at most 3, with a probability of at least 0.80 of passing when cation exchange clears as its own data show.

Table 7.9: Number of cation exchange verification runs, from a simulation of the verification.

Attribute	Fold in the cation exchange data[a)]	Fold the criteria rely on	SD of the log fold	Runs	Probability of passing		Runs needed[b)]
					Fold as in the data	Fold equal to the required	
Aggregate	2.52	1.40	0.065	1	1.00	0.052	1
				2	1.00	0.049	
				3	1.00	0.050	
				4	1.00	0.050	
				5	1.00	0.051	
Fragments	1.56	1.30	0.116	1	0.47	0.050	3
				2	0.72	0.050	
				3	0.86	0.049	
				4	0.94	0.051	
				5	0.97	0.050	

[a)] At set-point, from the selected model of aggregate. Fragments, which have no adequate model, from the centre points.
[b)] The smallest number of runs, at most 3, with a probability of passing of at least 0.80 when cation exchange clears as its data show.
100,000 simulated verifications per number of runs. A verification passes when the one-sided 95 % lower confidence bound of the mean log fold is at least the fold the criteria rely on.

The two probability columns give the chance of passing when cation exchange clears as its own data show and when it clears exactly the factor the criteria rely on. Aggregate needs 1 run and fragments 3, because the clearance factor in the cation exchange data exceeds the required factor by much more for aggregate than for fragments, 2.52 against 1.40 and 1.56 against 1.30. The larger of the two needs decided the study, so 3 runs were made. Table 7.10 gives them.

Table 7.10: Cation exchange runs on material from the re-centred low-pH step (study S-CEX-VER).

Run	Aggregate, % HMW (SEC)			Fragments, % LMW (nrCE-SDS)		
	Load	Pool	Fold	Load	Pool	Fold
1	2.05	0.86	2.37	1.12	0.69	1.64
2	2.11	0.96	2.21	1.03	0.62	1.64
3	2.07	0.78	2.67	1.00	0.64	1.56
Geometric mean fold			2.41			1.61
Lower 95 % confidence bound[a)]			2.27			1.45
Fold the criteria rely on[b)]			1.40			1.30
Result			Passed			Passed

[a)] One-sided, of the mean log fold, with the run-to-run SD of the log fold from the cation exchange data, as in the simulation that set the number of runs.
[b)] The clearance factor through cation exchange that the intermediate acceptance criteria upstream of it assume (PCMP-001).
Load: material inactivated at pH 3.40, the re-centred set-point, for 120 min at 23 °C, the longest hold and the highest temperature of the NOR in force. Cation exchange at set-point. Fold: load divided by pool.

Both attributes passed. Cation exchange cleared aggregate by a clearance factor of 2.41 at the geometric mean of the three runs, with a lower bound of 2.27 against the 1.40 the criteria rely on, and fragments by a factor of 1.61 with a lower bound of 1.45 against 1.30. Every run cleared more than the required factor for both attributes. The cation exchange pool also met the criteria of that step, at most 0.96 % HMW against 5.00 % and 0.69 % fragments against 2.00 %. The material from the step at its new pH set-point therefore behaves at cation exchange as the criteria of this step assume.

7.5.3 Confirmation runs at the low end of the pH PAR

The revision of the hold-time NOR moves routine operation toward longer holds, which is the direction that adds aggregate and fragments. PCP-006 planned no study at the Version 2.0 NOR. Study S-VI-WC therefore held the pool at pH 3.2, the lower end of the range studied and of the pH PAR, for 120 min at 23 °C, the upper ends of the hold-time and temperature NORs in force. The hold-time and temperature PARs reach further than those two settings, to 180 min and 25 °C, so the runs bound what routine operation can deliver and not what the PAR allows. Each hold was followed by a cation exchange run at set-point. The study ran from 2026-09-26 to 2026-09-29, with 3 runs, as many as the cation exchange verification, and is registered as PD-006-03, the confirmation runs at the low end of the pH PAR.

Product quality met every criterion of the step. At manufacturing scale aggregate ran from 2.05 to 2.07 % HMW against 7.02 %, and acidic variants from 19.7 to 20.6

% against 18.0 to 40.0 %. Each of those verdicts holds with the offset at either end of its interval. Fragments, which have no measured offset, ran from 0.91 to 1.05 % at the scale of the laboratory model, against 2.60 %. Table 7.11 gives the cation exchange runs on that material and the decision on each clearance factor, by the rule of Section 7.5.2.

Table 7.11: Low-pH holds at the low end of the pH PAR, with hold time and temperature at the upper ends of their NORs, each followed by a cation exchange run (study S-VI-WC).

Run	Aggregate, % HMW (SEC)			Fragments, % LMW (nrCE-SDS)		
	Load	Pool	Fold	Load	Pool	Fold
1	2.09	0.87	2.42	1.01	0.70	1.45
2	2.07	0.78	2.66	0.91	0.66	1.39
3	2.07	0.87	2.38	1.05	0.78	1.35
Geometric mean fold			2.48			1.40
Lower 95 % confidence bound[a)]			2.33			1.25
Fold the criteria rely on[b)]			1.40			1.30
Result			Passed			Not passed

[a)] One-sided, of the mean log fold, with the run-to-run SD of the log fold from the cation exchange data, as in the simulation that set the number of runs.
[b)] The clearance factor through cation exchange that the intermediate acceptance criteria upstream of it assume (PCMP-001).
Load: material held at pH 3.20, the lower end of the pH range studied and of the pH PAR, for 120 min at 23 °C, the upper ends of the hold-time and temperature NORs in force. Cation exchange at set-point. Fold: load divided by pool.

For aggregate the rule is met. Cation exchange cleared aggregate by a clearance factor of 2.48 at the geometric mean, with a lower bound of 2.33 against the 1.40 the criteria rely on.

For fragments the rule is not met. Each of the three runs cleared more than the required factor, from 1.35 to 1.45 against 1.30, and the geometric mean was 1.40. The one-sided 95 % lower confidence bound of the mean log reduction value was 1.25, below the required factor, so the rule the simulation of Section 7.5.2 fixed before the runs is not met. The three runs are not pooled with the 3 runs of S-CEX-VER, which were made on different material, and no rule of the analysis was changed once the result was known.

The failure was accepted on its risk, and PD-006-03 records the grounds. Fragments in the pool leaving the hold reached 1.05 % at most, which already meets the drug substance criterion of 2.00 %, so the drug substance would meet that criterion even if cation exchange cleared no fragments at all. The risk of a lower fragment clearance at cation exchange therefore falls on the producer and not on the patient. Fragments

are tested by nrCE-SDS at release, so an excursion would be detected on the drug substance itself. The cation exchange pool of these runs also met the fragment criterion of that step, at most 0.78 % against 2.00 %. None of the 2,000 simulated batches of the integrated model falls outside the drug substance criterion for fragments, so that probability is below 0.15 % at one-sided 95 % confidence (Section 8). No criterion of this report rests on the failed bound. The fragment criterion of this step uses the platform worst case of cation exchange, not these runs (Section 7.1), and fragments at this step are judged on their measured results. The acceptance was decided after the result was known.

7.6 A-Mab concentration

A-Mab concentration was assessed one factor at a time, for the reasons Section 2 gives. Study S-VI-UNI ran the hold at 5 and 35 g/L in duplicate, with the other three parameters at their set-points as characterized. PCP-006 §6.4 required aggregate, acidic variants and the XMuLV log reduction on every run. The study measured aggregate and acidic variants only (PD-006-07), and the clearance was measured later in the confirmation study S-VI-UNI2 (PD-006-06). Table 7.12 gives both studies.

Table 7.12: A-Mab concentration at both ends of its range, in the study of the characterization campaign and in the confirmation study.

Study	A-Mab concentration (g/L)	Replicate	Aggregate, % HMW	Acidic variants, %	XMuLV LRF, log10
S-VI-UNI	5	1	1.87	20.3	n.t.
		2	1.93	20.3	n.t.
	35	1	1.93	20.3	n.t.
		2	1.88	20.3	n.t.
S-VI-UNI2	5	1	2.08	19.9	6.59
		2	1.98	20.6	6.82
	35	1	1.73	20.0	6.35
		2	2.02	19.8	6.32

Other parameters at their set-points as characterized. Aggregate and acidic variants at manufacturing scale. n.t., not tested. PD-006-07 records that S-VI-UNI measured no viral clearance. S-VI-UNI2 spiked XMuLV at both ends of the range.

For aggregate and acidic variants no effect of the concentration was detected within the range studied. In S-VI-UNI aggregate was 0.006 percentage points higher at 35 g/L than at 5 g/L, with a 95 % confidence interval from −0.155 to 0.166 points, and in S-VI-UNI2 −0.156 points, with an interval from −0.815 to 0.502 points. Both intervals contain zero, and both are narrow against the margin of 4.92 points between the highest single-batch prediction over the NOR and the criterion. At the scale of the laboratory model the two levels of S-VI-UNI differ by 0.012 and 0.017 points from the mean of the set-point runs of the qualification study, 1.94 %, whose own standard

deviation is 0.027 points. The acidic variants gave the same result at every run of S-VI-UNI, 20.6 %, which differs from the mean of the set-point runs of the qualification study by 0.10 points, inside the standard deviation of those runs, 0.33 points. Every result of both studies meets its criterion at manufacturing scale, aggregate at most 2.08 % against 7.02 % and acidic variants from 19.8 to 20.6 % against 18.0 to 40.0 %. The range 5–35 g/L is therefore supported for these two attributes.

The range is not supported for fragments, which neither study measured. PCP-006 §6.4 did not require fragments in the concentration study, the samples of S-VI-UNI were not retained, so no nrCE-SDS result can be obtained for them, and SEC is biased in this matrix (Section 3). For that attribute the class of the parameter rests on the cleavage reaction itself. Cleavage in the hinge region is a reaction of each molecule with the solvent, so the fraction of molecules cleaved does not depend on how many of them are present.

Study S-VI-UNI2 measured the XMuLV log reduction at both ends of the range, in duplicate, with the other parameters at their set-points as characterized. It ran from 2026-09-19 to 2026-09-29 and measured no fragments (Section 3). The log reduction was 6.71 $\log_{10}$ at 5 g/L and 6.34 $\log_{10}$ at 35 g/L, a difference of -0.37 $\log_{10}$ at the high end, with a 95 % confidence interval from -1.20 to 0.47 $\log_{10}$ against a run-to-run standard deviation of 0.39 $\log_{10}$ from the response-surface design ($p = 0.361$). No effect of the concentration on the inactivation was detected, which is what the hold acting on the envelope of the virus leads one to expect (Section 2). Four runs bound that effect loosely. The interval reaches 1.20 $\log_{10}$ on the side of a loss, which is more than the margin of 0.17 $\log_{10}$ that the claim has over the NOR, so these data alone cannot exclude a loss of clearance at the high end of the range that would matter. For XMuLV clearance the class of the parameter therefore rests on how the hold acts (Section 9). All four results lie inside the 95 % prediction interval of a single run at the set-points as characterized, which runs from 5.85 to 7.60 $\log_{10}$. Of the four, 3 lie below the claim of 6.60 $\log_{10}$. The claim is judged on the mean prediction of the model (Section 3.5), and a single spiked run carries the scatter of the infectivity assay.

7.7 Verification of the platform worst cases

The criteria of the steps upstream of this one assume that the low-pH hold adds no more aggregate, fragments and acidic variants than it added on the platform. Each assumption was checked on the response-surface models of this step, as PCMP-001 §7.5 requires. The check evaluates the model at every combination of NOR low, set-point and NOR high as characterized, which are the values the studies were planned against, so a later change of a NOR cannot move a criterion. It takes the largest increase the model predicts over those combinations and the one-sided 95 % prediction bound of a single batch at the least favourable of them. Every increase is taken against the aggregate level of the load in the process model at set-point, 1.55 % HMW, because the input pool of the designs was not tested (PD-006-08). Table 7.13 gives the three assumptions and the results.

Table 7.13: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign			Outcome
				At the edges	Within the NOR	95 % bound	
Low-pH viral inactivation	Aggregates, HMW	Increase, largest (points)	0.70	0.84	0.53	0.56	Better, widened[a)]
	Fragments, LMW		0.15	0.22	0.10	0.19	Confirmed
	Acidic variants		1.00	0.60	0.40	0.40	Confirmed, not widened[b)]

[a)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case. The criteria upstream of this step were recalculated with the bound.
[b)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case, but the model is not adequate by the criteria of PCMP-001: it fails a test, or it fits the results exactly and the tests cannot be applied. The platform value stands.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

The column headed at the edges holds the largest increase measured at the edges of the range studied, against the same reference load. The outcome column applies the definition of PCMP-001 §7.5, under which an assumption is confirmed when the largest increase predicted within the NOR is no larger than the platform value. That definition rests on the model of the attribute, and only the aggregate model is adequate, so the confirmations for fragments and acidic variants carry the limitation Section 11 states. Only aggregate moved a criterion.

For aggregate the platform value is an increase of 0.70 percentage points, the largest increase predicted within the NOR as characterized is 0.53 points and the bound of a single batch is 0.56 points. The bound is better than the platform value, and the selected model of aggregate meets every adequacy criterion of PCMP-001 §7.2. PCMP-001 §7.5 and PCP-006 §7.4 both require that of a model before its bound may replace a platform value. PCP-006 §7.4 also asks whether the bound depends on a single run. It does not. With each run that PCP-006 §7.1 flags for examination deleted in turn, the bound is at most 0.55 points, still below the platform value (Section 5.3.2). The bound therefore replaced the platform value, and the aggregate criteria of the steps upstream of this one were recalculated with it.

The criterion affected is the aggregate criterion of the production bioreactor output. Its a priori value is 9.3 % HMW. The cation exchange verification tightened it to 6.32 %, and the bound of this step then widened it to 6.46 %, which is stricter than the a priori value and wider than the value the cation exchange verification alone would give.

The platform values are increases over a hold of 180 min, and the bound of this step is the increase within a NOR whose longest hold is 120 min, so the replacement reflects the shorter hold of the A-Mab NOR and not a more stable product. At a hold of 180 min, the platform hold, A-Mab added up to 0.84 points, more than the platform value of 0.70 points. The aggregate model carries no pH term, and the NOR in force ends at the same longest hold and warmest temperature as the NOR as characterized, so

the same bound covers the NOR in force. A hold longer than the upper end of the NOR is a departure from the control strategy of the step, handled under the quality system of ICH Q10, and the criterion carries no allowance for one.

The widened criterion is an intermediate criterion at the production bioreactor, and the drug substance criterion for aggregate is unchanged, so the risk of the change falls on the producer and not on the patient. What the downstream steps must clear is the factor the criteria already credit, the aggregate clearance of cation exchange, which Section 7.5.2 verified on material from this step. Aggregate is tested against its drug substance criterion by SEC at release. No simulated batch of the integrated model falls outside that criterion, so the probability is below 0.15 % at one-sided 95 % confidence, and the capability index of the attribute is 15.7 (Section 8). PCR-003 argues the widened criterion and its risk for the bioreactor.

For fragments and acidic variants the platform values stand. For fragments the largest increase predicted within the NOR as characterized is 0.10 points against a platform value of 0.15 points, and at the edges of the range studied the step added up to 0.22 points, more than the platform value. The bound of a single batch is 0.19 points, which is not better than the platform value, so no replacement follows and the platform value stands in the fragment criterion of the Protein A output. At the rate of Section 7.5.1, a hold at the upper end of the NOR in force adds about 0.10 points. For acidic variants the largest increase predicted within the NOR is 0.40 points and the bound is the same value, both better than the platform value of 1.00 points, and every run at a hold of 180 min added less than the platform value. The bound still does not replace it, because the model of the attribute fits the results exactly and the adequacy criteria of PCMP-001 §7.2 cannot be applied to it (Section 5.3). The acidic variant criteria upstream of this step therefore stay at their a priori values.

8 Process capability and robustness

Capability at commercial scale comes from the integrated model of the train, which runs 2,000 simulated batches through every step with each parameter drawn within its NOR in force (PCMP-001 §6.5). Table 8.1 gives the four attributes this step sets or changes.

Table 8.1: Commercial-scale capability of the attributes this step sets or changes, from the integrated model.

Attribute	Criticality	Drug substance criterion	Mean	SD	Cpk	Against PCMP-001
Viral clearance — XMuLV (enveloped)	VH	≥ 16.7 log10 (cumulative)	19.9	0.624	1.70	Above target
Aggregates (HMW)	H	≤ 5 % HMW (SEC)	0.785	0.0897	15.66	Above target
Fragments (LMW, clipped species)	L	≤ 2 % (nrCE-SDS)	0.643	0.0426	10.62	Above target
Acidic charge variants (deamidation)	VL	18–40 % (CEX/icIEF)	22.4	1.39	1.05	Between minimum and target

Mean and SD (standard deviation) of the simulated batches in the unit of the criterion. Cpk against the nearer limit, the lower of the two one-sided indices for a two-sided criterion. Target and minimum from PCMP-001 §7.9.

Cumulative XMuLV clearance has a capability index of 1.70 against its floor of 16.7 $\log_{10}$, which is above the target of 1.33 (PCMP-001 §7.9). The simulated mean is 19.9 $\log_{10}$ with a standard deviation of 0.62 $\log_{10}$, and the three claimed steps together put the distribution far above the requirement. This step contributes the largest of the three claims (Section 7.1).

Of the attributes the hold can add to, acidic variants have the lowest capability index, 1.05, which lies between the minimum of 1.00 and the target of 1.33. PCMP-001 §7.9 calls a value between the two acceptable and asks for the side of the criterion that limits it and the parameters that drive it, which PCMR-001 consolidates. The criterion is two-sided and the index is limited by its lower end, 1.05 against 4.24 at the upper end. The low-pH hold only raises acidic variants, by 0.20 percentage points per 60 min (Section 5.4), so the step moves the attribute away from the limit that binds. A longer hold would raise the index, and no change of this step is proposed on that ground, because the index is set by the level the production bioreactor forms and by its batch spread. PCR-003 gives that analysis and the mitigation. Aggregate and fragments have indices of 15.7 and 10.6, far above the target. The drug substance mean of aggregate is 0.78 % against a limit of 5.00 %, and of fragments 0.64 % against 2.00 %, because cation exchange and the steps after it remove both.

The same simulation gives the probability that the drug substance is out of specification. No simulated batch falls outside the drug substance criterion for aggregate or for fragments, so each of those probabilities is below 0.15 % at one-sided 95 % confidence, a bound that covers the sampling of the simulation and not the uncertainty of the integrated model itself. For acidic variants 1 of 2,000 batches falls below the lower limit, and the tail probability of a normal distribution with the simulated mean and standard deviation is 0.08 %. For XMuLV clearance no batch falls below the requirement and the normal tail probability is 0.18 per million.

Three results bound the robustness of the step. The claim holds across the NOR on the selected model, with the margins of Table 7.6. The quality attributes met their criteria in every run of the designs, which span the region studied, and in the confirmation runs at the low end of the pH PAR with hold time and temperature at the upper ends of their NORs (Section 7.5.3). The one pool that was held met its criteria after 7 days at 21 °C and 25 °C (Section 7.3), so a hold of a pool of that composition before cation exchange within that window does not threaten the criteria of the next step.

9 Parameter classification

Each parameter was classified under SOP-4001 by the decision logic of PCMP-001 §7.7. A critical process parameter (CPP) has an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation. A well-controlled critical process parameter (WC-CPP) has the same kind of effect but a low risk of falling outside that range, because it is readily controlled and its PAR is wide against its NOR. A key process parameter (KPP) affects process performance and no CQA, and a general process parameter (GPP) affects neither. Neither PCMP-001 nor SOP-4001 sets a numerical threshold for an effect that is practically relevant or for a PAR that is wide against its NOR. Where a class turns on such a size, this report names the judgement and the basis it was taken on. The size of each effect across the PAR and the margin from the PAR to the NOR are given with each class. The classes are in Table 4.1.

- *Inactivation pH (CPP).* A rise in pH across the range studied lowers the XMuLV log reduction by 1.22 $\log_{10}$ ($p < 0.001$), which is 19 % of the claim. Across the PAR of pH 3.2–3.61 the same model gives a change of 0.63 $\log_{10}$, so the claim is lost at the upper edge of that range. The pH of a large acidified pool is set by acid addition and mixing, which is the least directly controlled of the three settings, and pH control at manufacturing scale holds about 0.10 pH units, the half-width of the NOR itself. The margin from the upper NOR edge to the PAR edge is 0.11 pH units, so a drift of the size the NOR allows about the set-point would reach the edge of the PAR. No plan sets the margin at which that risk becomes high, so the class rests on that comparison as a judgement, the same comparison the decisions of Section 7.5 were taken on. On it the risk is high and the parameter is a CPP.
- *Hold time (WC-CPP).* A longer hold raises the XMuLV log reduction by 1.21 $\log_{10}$ over the range studied ($p < 0.001$) and aggregate by 0.472 percentage points over the same range ($p < 0.001$). It also raises the acidic variants by 0.4 points over that range and the fragments by 0.10 points at the rate S-VI-OFAT measured (Section 7.5.1). Across its PAR of 84–180 min the clearance effect is 0.97 $\log_{10}$, and the quality attributes stay inside their criteria across the whole range studied. The effect on a CQA makes the parameter critical. The lower PAR edge lies 16 min below the NOR, a margin of 80 % of the width of the NOR. A hold time is counted by the control system from a defined start, and the largest error of a hold-time record in these studies was the 4 min between the two records of DEV-006-01, which is 4 times smaller than the margin. No plan sets the margin at which the risk becomes low, so the class rests on that comparison as a judgement. On it the risk is low and the parameter is a WC-CPP.
- *Temperature (WC-CPP).* A warmer hold raises the XMuLV log reduction by 0.57 $\log_{10}$ over the range studied ($p = 0.036$) and aggregate by 0.215 percentage

points ($p < 0.001$). Across its PAR of 17.2–25 °C the clearance effect is 0.45 $\log_{10}$. The effect on a CQA makes it critical. The lower PAR edge lies 2.8 °C below the NOR, a margin of 93 % of the width of the NOR, and the hold is held near ambient in a jacketed vessel, so the risk of falling outside the acceptable range is low on the same judgement as the hold time. The temperature term of the XMuLV model depends on run 5, and without that run the XMuLV model carries no temperature term (Section 5.3.2). The effect on aggregate does not depend on it. Temperature raises aggregate by 0.204 to 0.226 percentage points over the range studied in every model selected after a deletion, and significantly in each ($p < 0.001$ at its weakest), so the parameter affects a CQA whichever flagged run is left out.

- *A-Mab concentration (GPP).* No effect of the concentration was detected on aggregate, on acidic variants or on the XMuLV log reduction across the whole range studied, which is the whole NOR of the parameter (Section 7.6). For aggregate and acidic variants the concentration studies bound the effect well against the margins to the criteria. For XMuLV clearance they do not. Four runs put the difference at the high end of the range at -0.37 $\log_{10}$, with a 95 % confidence interval from -1.20 to 0.47 $\log_{10}$, which reaches further on the side of a loss than the margin of 0.17 $\log_{10}$ that the claim has over the NOR. Four runs cannot establish the absence of an effect of that size, so for that attribute the class rests not on the absence of a detected effect but on how the hold acts. The acid denatures the envelope of the virus, and the antibody around it neither shields the envelope nor takes part in that reaction (Section 2). Neither concentration study measured fragments, so for that attribute the class rests on the cleavage reaction (Section 7.6). On the evidence available the parameter affects no CQA and no performance attribute of the step, so it is a GPP.

The step therefore carries one CPP and two WC-CPPs. A WC-CPP is a critical process parameter with good control capability, so all three are critical and all three are controlled within the limits of the design space.

10 Contribution to the control strategy

The step contributes four things to the control strategy of the drug substance process, and PCMR-001 consolidates them with the other steps.

1. *The claim and the ranges that assure it.* The step claims 6.60 $\log_{10}$ of XMuLV clearance, which it meets across the NOR in force at the least favourable combination (Section 7.2). The ranges in force are pH 3.3–3.5, 100–120 min and 20–23 °C, each inside the PAR by the margin Table 7.6 gives. The set-points are pH 3.4, 110 min and 21 °C. Inactivation pH is controlled as a CPP and the other two as WC-CPPs (Section 9).
2. *Monitoring and testing.* XMuLV clearance cannot be measured in manufacture, so the claim rests on keeping inactivation pH, hold time and temperature within their NORs on every batch. A-Mab concentration is a general process parameter within 5–35 g/L (Section 7.6). Aggregate, fragments and acidic variants have drug substance criteria whose methods the CQA register names (Table 2.1), so an excursion that reached the drug substance would be found by the release tests of those attributes.
3. *What this step hands to the next.* The neutralized pool is the load of cation exchange, which removes the aggregate the hold forms. The criteria of this step credit cation exchange with a clearance factor of 1.40 for aggregate and 1.30 for fragments, and Section 7.5.2 verified both on material from the step at its new pH set-point. On the material of the confirmation runs the aggregate factor was verified again, and the rule set for the fragment factor was not met and was accepted on its risk (Section 7.5.3). One pool of the step, made at the set-points as characterized, was held for up to 7 days at 21 °C and 25 °C and stayed inside the criteria (Section 7.3). Cation exchange is characterized under PCR-007.
4. *The attributes other steps control.* The cumulative XMuLV requirement is met by this step together with anion exchange (PCR-008) and virus filtration (PCR-009). MVM clearance is claimed by anion exchange and virus filtration, and this step claims none, because a parvovirus has no lipid envelope for low pH to disrupt. Aggregate and acidic variants are formed in the production bioreactor (PCR-003) and fragments during harvest and clarification (PCR-004), and this step adds a bounded amount to each (Section 7.7). The criteria assume no change in host cell protein across this step, and no clearance of it is claimed here (PCR-005).

One lifecycle control follows from the verification. The bound that widened the aggregate criterion of the production bioreactor output covers holds up to 120 min at up to 23 °C (Section 7.7). A change that extended the hold-time or temperature NOR of this step beyond those ends would take routine manufacture outside the basis of that criterion, and is assessed under the change management system of ICH Q10 (International Council for Harmonisation 2008).

11 Discussion

The step is understood well enough to be operated and transferred. The three parameters trade against each other along the contour of Figure 5.4, and no interaction among them was resolved by 18 runs. Aggregate grows with the hold time and with the temperature, the acidic variants and the fragments with the hold time, and all three stay far inside their criteria across the region studied. What limits the step is not product quality but the clearance claim, which is why the design space of Section 6 is bounded by the clearance contour and why the ranges in force were set from it.

The ranges in force were decided on these data, after the results were seen (Section 7.5). The evidence that they are safe to operate in is of three kinds. The model predicts the claim at the least favourable combination of the NOR with a margin of 0.17 $\log_{10}$. The quality attributes were measured at the new pH set-point over hold times from 0 to 180 min (Section 7.5.1) and at the low end of the pH PAR with hold time and temperature at the upper ends of their NORs (Section 7.5.3), and every result met its criterion. Cation exchange was run on material from both of those conditions and cleared aggregate as the criteria assume.

Confidence in the scale-down model is highest for the two outputs the qualification compared at both scales. The offsets of aggregate and acidic variants are small, their intervals contain zero, and the conclusion over the NOR is the same at either end of both intervals, so the model is qualified for both (Section 3.1). Six limitations bound what this report claims.

1. *Fragments and viral clearance were not compared at both scales.* Their offsets are unknown rather than zero. For fragments the conclusion rests on the assumption that cleavage in the hinge region depends on pH, time and temperature and not on the size of the vessel, and the margin of 1.47 percentage points to the criterion bounds what the assumption can cost. For viral clearance it rests on the spiking model, in which the matrix, the pH and the temperature are those of manufacture and only the volume differs.
2. *The log reduction carries the variability of a cell-based assay.* Its coefficient of variation at the centre points is 5.7 %, the largest of any response. The claim is judged on the mean prediction (Section 3.5), and a single spiked run at the least favourable corner of the NOR can fall below it (Section 7.2). The screening and the response-surface design also gave different pH effects on the same corner settings (Section 5.2), which bounds how precisely one campaign of spiked runs fixes that effect.
3. *Two results depend on one run.* Both selected models meet every adequacy criterion, and no conclusion over the NOR changed when a flagged run was deleted (Section 5.3.2). The temperature term of the XMuLV model and the

width of the XMuLV PARs both depend on run 5, so the margins of Table 7.6 are the conservative reading of 18 runs rather than a result a larger design has confirmed. The class of temperature does not depend on that run, because its effect on aggregate holds in every model (Section 9).

4. *Two outputs have no adequate model.* Acidic variants and fragments are judged on their measured results and set no range (Section 7.2). For the acidic variants the reason is that the results carry no scatter, so the adequacy criteria cannot be applied, and the record gives no reason for that (Section 5.1). For fragments the reason is that the selected model predicts left-out runs much worse than it fits its own, which is a design of 18 runs against an increase about 1.6 times the repeatability of the method over the whole hold-time range studied. Neither attribute constrains the region of the step on the evidence available, and neither claim extends beyond the measured results. The verification of the fragment and acidic variant platform worst cases rests on those models, so neither confirmation is as strong as the aggregate one (Section 7.7).
5. *Three approaches were set after the results they judge.* The model selection of corrective action CA3 was adopted on 2026-09-15, after the designs had been analysed. The approach to the hold study (Section 7.3) and the acceptance of the fragment bound of the confirmation runs (Section 7.5.3) were both decided once those results were known. Two rules were fixed before the results they decide, the clearance factors and the run counts of the cation exchange verification runs and of the confirmation runs (Section 7.5.2).
6. *The input pool of the designs was not tested.* The increase across the step that the verification needs is taken against the aggregate level of the load in the process model at set-point rather than against a measured load (PD-006-08).

The record of this report holds two study events and eight discrepancies from the plan (Section 13). Five of the discrepancies are studies made during the re-analysis, each with what was planned, what was done and why.

12 Conclusions

1. The step meets its claim of 6.60 $\log_{10}$ of XMuLV clearance across the NOR in force, which is pH 3.3–3.5, 100–120 min and 20–23 °C. The lowest mean predicted log reduction in that NOR is 6.77 $\log_{10}$.
2. The proven acceptable ranges are pH 3.2–3.61, 84–180 min and 17.2–25 °C. One edge of each is set by the clearance claim and the other is the end of the range studied, and each contains the NOR in force. Of the region studied, 63.9 % meets the criteria of the two outputs that have an adequate model.
3. Aggregate, fragments and acidic variants meet their intermediate acceptance criteria across the NOR in force, with the margins of Table 7.2, and the neutralized pool still meets them after 7 days at 21 °C and 25 °C. On material from the confirmation runs cation exchange cleared aggregate as the criteria assume and did not meet the rule for fragments, which was accepted on the risk PD-006-03 records (Section 7.5.3).
4. The scale-down model is qualified for aggregate and for acidic variants. Fragments and viral clearance were not compared at both scales. Their conclusions rest on the spiking model and on the cleavage reaction, as Section 3.1 sets out.
5. Inactivation pH is a CPP, hold time and temperature are WC-CPPs and A-Mab concentration is a GPP.
6. Within the NOR as characterized the step adds less aggregate than the platform worst case at the bound of a single batch, which widened the aggregate criterion of the production bioreactor output from 6.32 to 6.46 % HMW. The platform values for fragments and acidic variants stand.
7. The tightest capability of the attributes this step changes is the acidic variant index of 1.05 against the lower end of its criterion, which the low-pH hold moves away from. Cumulative XMuLV clearance has an index of 1.70.
8. The record holds two study events and eight discrepancies from the plan. These ranges, classes and controls roll up into PCMR-001.

13 Study events and discrepancies from the plan

PCP-006 calls a departure from the plan a deviation. A process characterization study is not manufacture under GMP, so this report records the events of execution as study events, under the numbers Version 1.0 gave them, and every departure from the plan as a discrepancy from the plan.

13.1 Study events

The record of the studies holds two events of execution. Table 13.1 lists them, and each is described below with its magnitude and its outcome.

Table 13.1: Study events recorded during the studies of this step.

Event	Summary	Detected during	Outcome
DEV-006-01	Hold-time record discrepancy between DCS and manual log	batch-record reconciliation	run retained in the analysis
DEV-006-02	XMuLV infectivity assay cytotoxicity at low dilution	assay review	run retained in the analysis

DEV-006-01, hold-time record discrepancy between the control system and the manual log

During batch-record reconciliation the hold time recorded by the distributed control system (DCS) and the hold time in the manual log were found to differ by 4 min. The root cause recorded is an offset between the clocks of the two records. Whichever record is right, the impact is bounded by the effect of the hold time on each output. A hold-time error of that size changes the predicted XMuLV log reduction by 0.040 $\log_{10}$, which is smaller than the residual standard deviation of its model, 0.39 $\log_{10}$, and smaller than the margin of 0.17 $\log_{10}$ at the least favourable NOR combination. It changes the predicted aggregate by 0.016 percentage points against a margin of 4.92 points to the criterion. The record names no single run, so the event is assessed for the design as a whole. Every run was analysed at its design settings and retained in the analysis.

DEV-006-02, cytotoxicity of the XMuLV infectivity assay at low dilution

The assay review found that the sample matrix was cytotoxic to the indicator cells of the XMuLV infectivity assay (AMV-3017) at low dilution, with the cytotoxicity recorded at a dilution factor of 8. The root cause recorded is the matrix of the sample, and the record names no component of it. The record names no affected run either, so the event is assessed for the study as a whole. Wells that are cytotoxic cannot be read for infection, so the titre of an affected sample is read from higher dilutions, which raises the lowest titre the assay can report. A higher limit for the output titre can only make a computed log reduction smaller, never larger. Every run was retained in the analysis.

13.2 Discrepancies from the plan

The register holds eight discrepancies from the plan. A discrepancy is any departure from the plan this report states results against, whether work the plan did not foresee or a requirement of the plan that was not met. Table 13.2 gives what the plan foresaw, what was done and why for each of them.

Table 13.2: Discrepancies from the plan PCP-006.

Discrepancy	Subject	Planned	Done	Reason
PD-006-01	OFAT stability study at the re-centred inactivation pH	PCP-006: no study at the Version 2.0 set-points; the response-surface design covers pH 3.2 to 4.0	a low-pH hold at pH 3.40 against pH 3.50 as control, over hold times 0 to 180 min, in duplicate (study S-VI-OFAT)	Version 2.0 re-centres the pH and hold-time set-points to give the XMuLV claim a margin that large-scale pH control can hold; the study shows the quality attributes in scope at the new pH directly

Discrepancy	Subject	Planned	Done	Reason
PD-006-02	Small-scale cation exchange verification runs on material inactivated at pH 3.40	PCP-006: no downstream verification	cation exchange runs at set-point on the worst-case material of the re-centred step (study S-CEX-VER), the number of runs from a simulation	to confirm that cation exchange clears aggregate and fragments from that material at least as well as the Version 2.0 criteria assume

Discrepancy	Subject	Planned	Done	Reason
PD-006-03	Confirmation runs at the low end of the pH PAR, with hold time and temperature at the upper ends of their NORs, after the hold-time NOR was revised	PCP-006: no study at the Version 2.0 NOR	low-pH holds at the lower end of the pH PAR, the upper end of the hold-time NOR and the upper end of the temperature NOR, each followed by a cation exchange run at set-point (study S-VI-WC), as many runs as the cation exchange verification; the rule set before the runs for fragment clearance by cation exchange was not met, and that was accepted on its risk, because the hold output meets the drug substance fragment limit without that clearance and release testing measures fragments by nrCE-SDS	the hold-time NOR was revised to give the upper edge of the pH NOR a margin to the XMuLV PAR edge; the revision moves routine operation toward longer holds, so the runs confirm that the least favourable combination for the product (the low end of the pH PAR, hold time and temperature at the upper ends of their NORs) harms neither product quality nor cation exchange

Discrepancy	Subject	Planned	Done	Reason
PD-006-04	Model selection for the response-surface outputs	PCP-006: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report

Discrepancy	Subject	Planned	Done	Reason
PD-006-05	Hold study of the neutralized low-pH pool, made during the Version 2.0 re-analysis	PCP-006: no hold study; Version 1.0 of the plan did not foresee one	the neutralized pool held at two temperatures for seven days, aggregate by SEC, acidic variants by icIEF and fragments by nrCE-SDS at every time point (study S-VI-HLD); the report states how a held pool is judged, an approach no plan set, chosen after the results were known	Version 2.0 judges a held intermediate against its intermediate acceptance criteria, and Version 1.0 had no hold result to judge
PD-006-06	XMuLV clearance at the two ends of the A-Mab concentra-tion range	PCP-006: XMuLV clearance measured on every run of the A-Mab concentration study (study S-VI-UNI)	XMuLV-spiked low-pH holds at the low and the high end of the A-Mab concentration range, in duplicate, the other parameters at their set-points as characterized (study S-VI-UNI2)	the concentration study did not measure XMuLV clearance (PD-006-07), so the clearance across the concentration range had no result

Discrepancy	Subject	Planned	Done	Reason
PD-006-07	XMuLV clearance not measured in the concentration study	PCP-006: XMuLV clearance, aggregate and acidic variants measured at each A-Mab concentration of the univariate study	aggregate and acidic variants measured, XMuLV clearance not; the confirmation study S-VI-UNI2 measured it at the two ends of the range (PD-006-06)	virus spiking was scheduled for the designed studies only
PD-006-08	Input pool of the design runs not tested	PCP-006: the input pool of every design run tested for aggregate, acidic variants and A-Mab concentration	no input pool result is in the record; the increase across the step is taken against the load levels of the process model at set-point	the input samples were not scheduled in the study record

Five of them are studies made during the Version 2.0 re-analysis, after the results of the designs were known. They are PD-006-01, the study at the new pH set-point (Section 7.5.1), PD-006-02, the cation exchange verification runs (Section 7.5.2), PD-006-03, the confirmation runs at the low end of the pH PAR (Section 7.5.3), PD-006-05, the hold study of the neutralized pool (Section 7.3), and PD-006-06, the clearance at the two ends of the concentration range (Section 7.6). PD-006-04 is the model selection (Section 3.5), which INV-002 corrective action CA3 applies to every report of the campaign in place of the full quadratic model the plan prescribed, and the comparison of the models within 2 AICc units of each selected model is registered within it (Section 5.3.1). The remaining two are requirements of the plan that were not met, PD-006-07 and PD-006-08, and the two subsections below describe them.

What the model selection changed at this step is the following. The aggregate criterion of the production bioreactor output is 6.46 % HMW with the selection and 6.19 % without it, because the verification bound of the cation exchange aggregate clearance comes from the selected model of that step (Section 7.7). The aggregate criterion of this step is 7.02 % with the selection and 6.75 % without it, for the same reason. The acidic variant criterion of the production bioreactor output is 39.0 % with the selection and 39.6 % without it. The comparison without the selection takes the bound of the full quadratic model of the acidic variants of this step, which reproduces its results exactly. An exact fit is never adequate, no model of that output

of any form is adequate, and no bound of this step therefore replaces that platform value (Section 7.7). The bound this step contributes is nearly the same under either model of aggregate, 0.56 percentage points on the selected model and 0.56 points on the full quadratic. Every conclusion over the NOR, every proven acceptable range and both scale-down qualifications of this step are the same under the criteria the full quadratic models would have given.

PD-006-07, XMuLV clearance not measured in the concentration study

PCP-006 §6.4 required aggregate, acidic variants and the XMuLV log reduction on every run of the concentration study S-VI-UNI. The study measured aggregate and acidic variants only, because spiking had been scheduled for the designed studies alone and no titre was taken in the concentration study. The departure was found when the plan was compared with the study record during the Version 2.0 re-analysis, and Version 1.0 of this report did not notice it. The consequence was that the clearance across the concentration range had no result, while the range 5–35 g/L was supported for aggregate and acidic variants alone. The confirmation study S-VI-UNI2 then measured the clearance at the two ends of the range in duplicate (PD-006-06). It detected no effect of the concentration, and with four runs it cannot exclude a loss at the high end of the range larger than the margin of 0.17 $\log_{10}$ that the claim has over the NOR (Section 7.6). For XMuLV clearance the class of the parameter therefore rests on how the hold acts and not on the absence of a detected effect (Section 9).

PD-006-08, input pool of the design runs not tested

PCP-006 §5.3 and §5.4 require a sample of the input pool, taken before acidification, on every run of both designs, tested for aggregate, acidic variants and A-Mab concentration. No input pool result is in the record of either design, because the input samples were not scheduled in the study record. The departure was found during the Version 2.0 review of the study record against PCP-006, after the designs had been analysed. The consequence is that the increase in an attribute across the step cannot be formed run by run. The verification of the platform worst cases needs that increase, and it is therefore taken against the aggregate level of the load in the process model at set-point, 1.55 % HMW (Section 7.7). The A-Mab concentration of the design runs is likewise the value set and not a measured result (Section 3). Every criterion of the step is on the output of the step, so no judgement of an output against its criterion is affected, and every run was retained in the analysis.

Appendix A. Screening design matrix and responses

Table A.1 gives the 11 runs of the screening design in natural units, with every result of the neutralized pool measured on them. The input pool was not tested (PD-006-08). The fragment results of this design are by SEC and are reported for information only (Section 3).

Table A.1: Low-pH viral inactivation screening design, settings and results.

Run	Type	Factor settings			Responses			Fragments, LMW
		A	B	C	HMW	XMuLV LRF	Acidic	SEC
		pH	min	°C	%	log10	%	%
1	factorial	3.2	60	15	1.69	7.02	20.5	1.43
2		3.2	60	25	1.90	7.48	20.5	1.44
3		3.2	180	15	2.16	8.17	20.9	1.42
4		3.2	180	25	2.32	9.37	20.9	1.57
5		4	60	15	1.69	4.81	20.5	1.43
6		4	60	25	1.90	5.14	20.5	1.37
7		4	180	15	2.14	6.62	20.9	1.41
8		4	180	25	2.44	7.17	20.9	1.50
9	center	3.6	120	20	2.01	7.35	20.7	1.41
10		3.6	120	20	2.07	6.92	20.7	1.45
11		3.6	120	20	2.01	7.36	20.7	1.53

Factors: A, inactivation pH. B, hold time. C, temperature.
LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001).
Not tested in this design: LMW by nrCE-SDS.

Appendix B. Response-surface design matrix and responses

Table B.1 gives the 18 runs of the response-surface design in natural units, with every result of the neutralized pool measured on them. The input pool was not tested (PD-006-08). The fragment results by nrCE-SDS are those of the retained samples, tested in study S-VI-RET, and they are the fragment results this report judges.

Table B.1: Low-pH viral inactivation response-surface design, settings and results.

Run	Type	Factor settings			Responses			Fragments, LMW	
		A	B	C	HMW	XMuLV LRF	Acidic	SEC	nrCE-SDS
		pH	min	°C	%	log10	%	%	%
1	factorial	3.2	60	15	1.69	6.49	20.5	1.37	0.96
2		3.2	60	25	1.88	7.35	20.5	1.39	0.89
3		3.2	180	15	2.17	7.65	20.9	1.42	0.98
4		3.2	180	25	2.36	7.91	20.9	1.52	1.13
5		4	60	15	1.68	4.82	20.5	1.45	0.93
6		4	60	25	1.88	5.77	20.5	1.45	0.87
7		4	180	15	2.11	6.45	20.9	1.48	0.99
8		4	180	25	2.39	7.16	20.9	1.52	1.03
9	axial	3.2	120	20	2.01	7.04	20.7	1.45	0.89
10		4	120	20	2.02	6.13	20.7	1.43	0.87
11		3.6	60	20	1.79	6.65	20.5	1.48	1.01
12		3.6	180	20	2.24	7.96	20.9	1.52	1.07
13	center	3.6	120	15	1.92	7.22	20.7	1.43	0.98
14		3.6	120	25	2.13	7.29	20.7	1.49	0.98
15		3.6	120	20	2.02	6.94	20.7	1.46	1.03
16		3.6	120	20	1.98	6.15	20.7	1.48	1.05
17		3.6	120	20	2.02	6.97	20.7	1.43	1.05
18		3.6	120	20	2.00	6.78	20.7	1.41	0.97

Factors: A, inactivation pH. B, hold time. C, temperature.
LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001).

Appendix C. Full effect and coefficient tables

Table C.1 gives every screening effect on aggregate and on the XMuLV log reduction. The screening model of the acidic variants reproduces its results exactly, with a hold-time effect of 0.4 percentage points and no other effect, so its table carries no further information. Table C.2 gives the coefficients of the selected model of aggregate, of the XMuLV log reduction and of fragments, whose model is not adequate, sets no range and is reported for completeness.

Table C.1: Screening effects on aggregate and on the XMuLV log reduction, every term.

Response	Term	Effect	Coef.	Std. err.	t	p-value	Sig.
Aggregates (HMW, %)	B	0.4715	0.2357	0.0118	20	< 0.001	***
	C	0.2209	0.1105	0.0118	9.37	< 0.001	***
	AC	0.03501	0.0175	0.0118	1.48	0.212	
	AB	0.02848	0.01424	0.0118	1.21	0.294	
	A	0.02209	0.01105	0.0118	0.937	0.402	
	BC	0.01069	0.005345	0.0118	0.453	0.674	
XMuLV LRF ($\log_{10}$)	A	−2.077	−1.038	0.0936	−11.1	< 0.001	***
	B	1.723	0.8616	0.0936	9.2	< 0.001	***
	C	0.6359	0.318	0.0936	3.4	0.027	*
	BC	0.2385	0.1192	0.0936	1.27	0.272	
	AB	0.2028	0.1014	0.0936	1.08	0.340	
	AC	−0.1939	−0.09696	0.0936	−1.04	0.359	

Effect, the change in the response over the coded range of the factor. Terms: A, inactivation pH. B, hold time. C, temperature. Effects of every term, largest first.

Table C.2: Coefficients of the selected response-surface model of each output, in coded units.

Output	Term	Coef.	Std. err.	t	p-value	Sig.
Aggregates (HMW, %)	Intercept	2.016	0.00428	471	< 0.001	***
	B	0.2362	0.00574	41.2	< 0.001	***
	C	0.1074	0.00574	18.7	< 0.001	***
XMuLV LRF ($\log_{10}$)	Intercept	6.818	0.0919	74.2	< 0.001	***
	A	−0.6119	0.123	−4.96	< 0.001	***
	B	0.6046	0.123	4.9	< 0.001	***
	C	0.2856	0.123	2.32	0.036	*
Fragments (LMW, %, nrCE-SDS)	Intercept	1.002	0.0179	55.8	< 0.001	***
	A	−0.01527	0.0149	−1.03	0.324	
	B	0.05404	0.0149	3.63	0.003	**
	A^2	−0.09646	0.0267	−3.61	0.003	**
	B^2	0.06188	0.0267	2.31	0.038	*

Terms: A, inactivation pH. B, hold time. C, temperature. A^2 and B^2 are the squared terms. The model of fragments is not adequate by PCMP-001 §7.2 and sets no range. The acidic variants are left out, because their model reproduces its results exactly.

Appendix D. Analytical methods summary

Table D.1 gives the method behind each output of the step, the sample it was measured in and its reportable unit. Table 3.3 gives which method measured fragments in which study.

Table D.1: Analytical methods of the outputs of the low-pH viral inactivation step.

Output	Sample	Technique	Method validation	Reportable
XMuLV log reduction factor	Spiked pool at the start and at the end of the hold	Infectivity titre (TCID50)	AMV-3017	log10 reduction
Aggregates, HMW	Neutralized pool	Size exclusion chromatography	AMV-3011	% HMW
Fragments, LMW	Neutralized pool	Non-reduced CE-SDS	AMV-3020	% LMW
Acidic variants	Neutralized pool	Imaged capillary isoelectric focusing	AMV-3013	%

TCID50, the 50 % tissue culture infectious dose. The log reduction factor is the difference between the log10 virus load at the start of the hold and the log10 virus load at the end, in the same spiked run. Fragments were measured by size exclusion chromatography before the method decision AMD-001 and by non-reduced CE-SDS from that date.

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